Selective Technological Disruption

Evidence from TV Entry and Radio Stations

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Abstract

When a new technology erodes an established source of revenue, firms may respond by expanding other parts of their business. I study this adjustment through the introduction of television and its effects on American radio stations. Using newly assembled station-level financial records and variation in television rollout associated with the 1948–1952 FCC licensing freeze, I find that television reduced annual major-network revenue by $8,634 per station while increasing local advertising revenue by $10,770. The estimated effect on total time-sales revenue is small but imprecise. These changes were concentrated among stations with preexisting major-network revenue. Stations with greater initial network dependence experienced larger network losses without correspondingly larger local advertising gains. Local advertising gains were also smaller in markets with competitors that received no major-network payments, even after accounting for population. The findings suggest that the ability to replace lost business varies with existing revenue sources and local competition.

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1 Introduction

Technological change can take away an important part of a firm's business without forcing the firm to close. A firm may lose some customers and continue serving others, or expand activities in which the older technology remains useful. Whether it can replace the lost revenue depends on what it was doing before competition arrived and the opportunities available elsewhere in its business. The American radio industry faced such a change with the arrival of commercial television. In the station financial records I assemble, annual payments from broadcasting networks fell from approximately $85 million in 1946 to $5 million in 1960, a decline of 94 percent in constant 1950 dollars. Radio stations continued to sell airtime, however, and advertising sold directly to local businesses became an increasingly important source of revenue. This raises a broader question about technological competition: how do firms respond when a new technology reduces the value of one activity, and what determines whether they can make up the lost business elsewhere?

I study this question using the introduction of commercial television and its effects on American radio stations. Before television, the four major national radio networks, ABC, CBS, NBC, and MBS, assembled programs and sold the advertising carried within them to national sponsors. Affiliated stations received payments for broadcasting these sponsored programs. Stations also sold airtime independently of the networks, to national and regional advertisers and to businesses in their own markets (Federal Communications Commission, 1941; Midgley, 1948; Head, 1956). These sources of revenue depended on different commercial relationships. A network affiliate received payments for programs and advertising supplied by the network, while its own advertising business depended on the customers it could attract and the airtime it could sell. Stations differed considerably in how much of their revenue came from each source.

Television changed the value of these relationships. The companies operating the major national radio networks were also involved in television, and programs, performers, and advertisers increasingly moved to the new medium (Head, 1956; Library of Congress, 2026, n.d.a). A radio station could therefore lose payments from a network as the network expanded its television operations. Television also competed for advertising that stations sold directly, particularly from national advertisers seeking large audiences. Local advertising presented a different possibility. Radio stations already served businesses interested in reaching customers within particular markets, and the departure of network programs could free airtime for programming and advertising controlled by the stations themselves. Television could reduce demand for this advertising as well, so the direction of the local revenue response was not predetermined. The question is whether radio stations expanded their own advertising

business as their network relationships weakened, and whether the response depended on their initial network dependence and the other stations with which they competed.

The radio-television transition is useful for studying these questions for two reasons. First, the Federal Communications Commission (FCC) collected financial accounts that separately recorded payments from national networks, advertising sold to national and regional customers, and advertising sold to local businesses. These records make it possible to follow the sources of revenue within individual stations, alongside operating expenditures and profit. Second, the expansion of television was interrupted by an FCC licensing freeze between 1948 and 1952. The interruption affected when television became available in different markets and provides a source of variation in the timing of competition from the new medium.

The main empirical difficulty is that television entered larger and more commercially attractive markets earlier (Gentzkow, 2006; Koenig, 2023). On September 30, 1948, the FCC suspended the processing of television-station applications while reconsidering technical standards and channel allocations. Proposed stations that had already received construction permits could continue toward operation, while applicants still awaiting authorization could not obtain permits until licensing resumed in 1952 (Federal Communications Commission, 1950, 1952; Radio News Bureau, 1949; Koenig, 2023). Television stations consequently began operating in some markets during the freeze, while entry in others was delayed. The freeze interrupted a process in which broadcasters had already applied to establish television stations, allowing me to compare radio markets placed on different paths to television availability. Construction-permit status was not randomly assigned, and the early and delayed markets differed in their pre-television characteristics. The empirical design accounts for these differences and uses pre-treatment outcomes to assess the comparison.

I construct a longitudinal panel of commercial radio stations using FCC financial records and contemporaneous editions of the *Broadcasting Yearbook*. The data contain station-level financial observations between 1946 and 1960, including payments from major networks, national and regional advertising revenue, local advertising revenue, operating expenses, and profit. I combine these records with television construction permits, broadcast dates, and signal coverage to identify when television first became available in each radio market. I also construct pre-television market characteristics from the 1940 U.S. Census. The preferred revenue sample contains 769 radio stations and 6,861 station-year observations.

I use two complementary difference-in-differences designs. The first compares stations in markets whose first television service came from broadcasters authorized before the freeze with stations in markets whose prospective television entry was delayed. This benchmark specification includes station and year fixed effects and allows predetermined market characteristics to have year-specific effects. My preferred analysis retains the freeze-based research-

design sample but uses the realized year of first local television exposure as the treatment date. I estimate these staggered effects using the imputation estimator of Borusyak et al. (2024), which allows treatment effects to differ across cohorts and years since entry, and report estimates using Callaway and Sant'Anna (2021) as an alternative. The identifying assumption is that, absent television, stations exposed at different times would have followed comparable outcome trends under the maintained specification. I examine pre-treatment trends and the differences in market characteristics that existed before the freeze.

I find that television entry reduced radio stations' revenue from national networks while increasing the advertising they sold to local businesses. In the preferred specification, annual major-network revenue falls by $8,634 per station, or 31.5 percent of its pre-television mean, and local advertising revenue increases by $10,770, or 12.5 percent. Both estimates are statistically significant at the 5 percent level. The estimated effect on total time-sales revenue is a decline of only $756, with a standard error of $7,777. The local advertising gain therefore approximately offsets the decline in network revenue, although the total-revenue estimate is too imprecise to rule out economically important changes in either direction. The benchmark freeze comparison gives a similar pattern for local advertising and total time-sales revenue. The long decline in national network broadcasting was accompanied by a different response in the advertising business that radio stations operated themselves.

The response was not the same across stations. Those that earned major-network revenue before television entered experienced larger local advertising gains than stations with no initial major-network revenue. Among network-connected stations, local advertising gains were larger in the lowest than in the highest quartile of initial network dependence, although they did not decline steadily across quartiles. The composition of competitors also mattered for the observed response. Network-connected stations in markets containing stations with no major-network revenue experienced smaller local advertising gains than network-connected stations in markets without such competitors. This difference remains statistically significant after accounting for 1940 market population. These comparisons show that television's effects varied with stations' initial revenue sources and the types of competitors in their markets. They do not separately identify the causal effect of market structure itself.

To understand these findings, I develop a framework in which national networks supply sponsored programming and radio stations decide how to use the remaining broadcast time. As television becomes available locally, a network may reduce the bundle of programs and advertisements it supplies to an affiliate. The station then loses network payments but gains time for its own programming and advertising sales. It cannot simply replace network programs with commercials, since advertising needs programming that attracts listeners. Television also competes for advertisers and may lower the rates stations can charge.

However, local businesses may have less reason than national and regional advertisers to shift toward television, and station-controlled programming may be particularly valuable for reaching local customers. The model shows how stations can expand local advertising sales enough to increase revenue despite lower advertising rates. These gains need not fully replace lost network payments, nor does every station have to lose network business. The framework provides a possible explanation for the revenue changes, although the financial records do not separately identify advertising quantities, prices, or programming decisions.

This paper contributes to research on how firms change their activities when competition alters the value of what they produce or sell. Bernard et al. (2010) document the creation and destruction of products within manufacturing firms, while Bernard et al. (2011) and Mayer et al. (2014) examine how trade liberalization and market competition affect firms' product mix. A related literature studies technological change through innovation, productivity, entry, exit, and the movement of economic activity across producers (Klette and Kortum, 2004; Collard-Wexler and De Loecker, 2015; Igami, 2017; Garcia-Macia et al., 2019). I study a setting in which competition from a new technology changes the value of separate commercial relationships within existing firms. Radio stations received payments for carrying national network programming and sold advertising through their own sales operations. The FCC accounts identify these revenues separately, allowing me to measure the loss of network business, the response of station-sold advertising, and the extent to which the two changes offset each other. The estimates also show how the response varied with stations' initial dependence on the business threatened by television.

The paper also relates to research on how established industries respond to competing technologies. Adner and Snow (2010) describe how firms using an older technology may concentrate on market segments in which that technology retains an advantage. In media markets, George and Waldfogel (2006) find that competition from the nationally distributed *New York Times* led local newspapers to place greater emphasis on local coverage. Angelucci et al. (2024) study the effects of television on newspapers and find declines in readership and advertising revenue, accompanied by reductions in newspaper content, particularly local news. Djourelova et al. (2025) examine newspapers' responses to Craigslist and show that initial dependence on classified advertising mattered for their responses to online competition. These studies establish that competition from a new medium can affect the products and customers of existing media firms. In radio, the financial relationship between local stations and national networks creates an additional margin. Television offered the networks another outlet for sponsored programming, while affiliates retained their own advertising businesses. I measure the revenue changes on each side of that relationship and examine how they differed across stations.

Finally, the paper contributes to the literature on competition within radio markets. Berry and Waldfogel (1999) study radio entry and welfare, while Berry and Waldfogel (2001) examine the relationship between ownership concentration and product variety. I examine how the arrival of a competing medium affected station revenues in radio markets containing different combinations of network-connected and other stations. The results document differences in local advertising responses across these markets, conditional on pre-television population, and provide evidence on how the competitive environment was associated with stations' responses to television.

Section 2 describes commercial radio, the development of television, and the FCC licensing freeze. Section 3 develops the conceptual framework. Section 4 describes the data, and Section 5 presents the empirical strategy. Section 6 reports the results, followed by the conclusion.

2 Historical Background and Institutional Setting

2.1 Commercial Radio Before Television

Commercial radio in the 1940s was organized around national broadcasting networks and the stations that carried their programs. The networks assembled entertainment, news, and other programs for simultaneous broadcast across affiliated stations, selling the advertising carried within them to national sponsors. Local stations received payments for carrying these programs, but they also operated their own schedules and sold advertising independently. By the end of the decade, four networks dominated national radio broadcasting: the National Broadcasting Company (NBC), the Columbia Broadcasting System (CBS), the American Broadcasting Company (ABC), and the Mutual Broadcasting System (MBS).

The network system had developed over the preceding two decades. NBC was established by the Radio Corporation of America (RCA) in 1926 and initially operated two networks, known as the Red and Blue Networks. CBS followed soon afterward and became NBC's principal national competitor. Mutual began in 1934 as a cooperative arrangement among several prominent stations, including WOR in New York and WGN in Chicago. ABC emerged from NBC's former Blue Network, which RCA sold in 1943 after the FCC adopted regulations restricting common ownership of national radio networks (Federal Communications Commission, 1941; Library of Congress, 2026; Head, 1956). The four networks differed in ownership and organization, but each relied on affiliated stations to distribute its programs and advertising across the country.

For affiliates, network programming provided both content and revenue. Networks sold

advertising to sponsors and supplied the resulting sponsored programs to stations, which received compensation for the airtime they provided. The amount of time committed to network programming and the terms of compensation differed across affiliation agreements (Federal Communications Commission, 1941; Midgley, 1948). The FCC financial accounts record these payments as revenue from the sale of station time to networks. I refer to payments from ABC, CBS, NBC, and MBS as *major-network revenue*. They are payments received by radio stations from the networks, distinct from the advertising revenue that networks collected directly from their sponsors.

Radio stations also sold airtime outside their network arrangements. National and regional advertisers purchased time on individual stations, often through representatives who arranged advertising across several markets. Local businesses bought time to reach listeners in their own cities and surrounding areas (Midgley, 1948). These purchases were known as spot advertising. The FCC distinguished advertising sold to national and regional customers from advertising sold to local customers, and recorded both separately from network payments. Stations thus earned revenue from three principal sources: national and regional spot advertising, local advertising, and payments from networks. The importance of each differed widely. Some stations received substantial payments from the major networks, while others earned little or no major-network revenue and relied primarily on advertising they sold themselves.

The arrival of television threatened the national network system in a particular way. The companies that dominated national radio were also among those developing television. NBC and CBS had experimented with the medium before World War II and expanded their television operations after the war. ABC began developing its own television network during the late 1940s (Library of Congress, 2026, n.d.a; Head, 1956). Mutual announced plans for a television network in 1950 but never established a lasting national operation (Broadcasting, 1950). The DuMont Television Network, developed by the television equipment manufacturer DuMont Laboratories, competed with NBC, CBS, and ABC before ceasing operations in the 1950s (Library of Congress, n.d.b). Television was therefore being developed by the organizations supplying radio stations with programming and by firms competing to enter national broadcasting.

The consequences for network radio were anticipated early. In April 1949, Merlin H. Aylesworth, NBC's founding president, published an article in *Look* titled "Radio Is Doomed." He predicted that "within three years the broadcast of sound, or ear radio, over giant networks will be wiped out" (Aylesworth, 1949; Broadcasting, 1949). His prediction concerned national network broadcasting, whose programs, performers, and sponsors were increasingly moving toward television. For the networks, television offered another way to distribute

sponsored programs and reach national audiences. For radio affiliates, the same movement could mean losing the payments and programming they had previously received from the networks. Yet those payments were only part of the station's business. Affiliates could continue selling advertising directly to customers in their own markets.

Some stations were already emphasizing this part of their business by the mid-1950s. In a 1954 advertisement, WAVE, an NBC affiliate in Louisville, promoted its news, sports, and local programming with the statement that "network attractions are only part of our appeal" (WAVE Radio, 1954). Richard Klaus, manager of WERE in Cleveland, drew a more explicit distinction between radio and television in 1955: "You might say TV is following the national magazine pattern. We pattern ourselves as closely as possible after the successful local newspaper" (Time, 1955). The same *Time* report described stations relying increasingly on recorded music, news, sports, and short advertising announcements in place of expensive network-style entertainment. The examples describe the approaches of individual stations, including one that continued to carry network programming. The financial records provide a broader account of how the composition of radio revenue changed during these years.

Figure 1 reports the composition of commercial radio time-sales revenue between 1946 and 1960. I divide revenue into network payments, national and regional advertising, and local advertising. Network payments include sales of station time to the four major networks and to regional and other network arrangements. Major-network payments accounted for approximately 94 percent of network revenue in the pre-television sample. Regional and other network payments together represented only about 1.3 percent of time-sales revenue. I therefore use major-network revenue to measure stations' initial financial dependence on the national broadcasting networks.

The change in revenue composition was large. In 1946, network payments accounted for approximately 22 percent of total station-time revenue; by 1960, their share was below 2 percent (Figure 1, Panel A). Local advertising moved in the opposite direction, rising from about 47 percent to 67 percent. The national and regional advertising share changed much less. Panel B restricts the sample to stations founded before 1949. Their network share also declined, while local advertising rose to approximately 61 percent of time-sales revenue by 1960. The decline in network dependence was therefore present among stations that had entered the industry before television became widely available, although the set of operating stations is not held fixed across years.

The corresponding revenue levels in Figure 2 show that network receipts declined in dollars as well as in revenue share. Local advertising followed a different path. Average local advertising revenue fell during the immediate postwar years but subsequently recovered, particularly among stations founded before 1949. The increasing local share was thus not

simply the result of network revenue falling while local revenue remained fixed. These national trends reflect several changes occurring at once, including the entry and exit of radio stations and changes in advertising conditions. They describe the industry's movement away from network payments but do not establish how much of that movement television caused.

2.2 The Rise of Television and the FCC Freeze

Commercial television expanded after World War II, although the opening of a television station required federal authorization. Broadcasters first applied to the Federal Communications Commission (FCC) for a construction permit. Once approved, the permit allowed an applicant to build a station and proceed toward operation. Television broadcasting was subject to technical restrictions on frequencies, signal interference, and the allocation of channels across cities. As applications increased during the late 1940s, the FCC faced difficulties accommodating additional stations within the existing system of channel assignments (Federal Communications Commission, 1950).

On September 30, 1948, the FCC suspended the processing of television-station applications while reconsidering its engineering standards and channel allocations. The order, which became known as the television freeze, stopped the processing of applications already pending before the Commission as well as newly submitted applications (Federal Communications Commission, 1950). The suspension was initially intended to be temporary. It lasted almost four years.

The freeze did not prevent stations that had already been authorized from beginning operation. A broadcaster holding a construction permit before September 30, 1948 could continue building its station and begin broadcasting once construction was completed. Applicants still awaiting authorization could not obtain construction permits during the suspension. Both groups had expressed an intention to enter television before the freeze, but only the first could proceed through the regulatory process. Television stations therefore continued to open in some markets while proposed stations in others remained unable to begin construction.

The distinction affected the timing of television availability. Markets reached by stations with existing construction permits could receive television during the freeze, while markets depending on pending applications had to wait until the FCC resumed processing. Permits did not determine the exact date of entry, however. Authorized stations still needed to complete construction, and television signals could reach areas outside the city in which a station was licensed. The regulatory status of an applicant and the date at which a radio market first received television were consequently related but separate measures.

The freeze ended on April 11, 1952, when the FCC adopted its Sixth Report and Order. The revised allocation plan added 70 Ultra High Frequency (UHF) channels to the 12 existing Very High Frequency (VHF) channels and expanded the number of communities assigned television channels. Application processing resumed on July 1, 1952 (Federal Communications Commission, 1952, 1953). Television stations could now obtain construction permits in markets where the suspension had delayed entry.

Figure 3 shows television's national expansion around the freeze. Panel A reports the cumulative number of stations on air, and Panel B reports new station openings by year. Entry increased after World War II and continued during the early years of the freeze as stations with existing construction permits completed construction. Few new stations opened during the later years of the suspension. Openings increased again after the FCC resumed processing applications in 1952. The freeze interrupted television's expansion at a time when broadcasters were already seeking permission to enter the industry.

I use the application and construction-permit records to identify the regulatory status of television stations when the freeze began, and subsequent operating dates and signal coverage to determine when television first became available in each radio market. Markets whose first signal came from a previously authorized station are distinguished from those whose first signal came from a station with an application pending in September 1948. This comparison exploits the difference in regulatory authorization at the start of the freeze, but permit status was not randomly assigned. Markets authorized earlier may have differed in characteristics that also affected radio revenue. Section 5 describes these differences, the comparison groups, and the assumptions required to identify the effects of television entry.

3 Conceptual Framework

A commercial radio station earns revenue in two ways. National networks supply sponsored programs, sell the advertisements that accompany them, and pay affiliated stations to broadcast them. Stations also sell advertising directly to local businesses and to national and regional advertisers. Unlike network payments, this revenue depends on the advertising time the station sells itself.

I consider one radio station in a local market. Television can make it less profitable for a national network to continue supplying sponsored programs to that station, even if the network continues broadcasting those programs elsewhere. The model has two stages. First, the network decides how much sponsored programming to provide as television becomes available locally. The station then chooses how to use the time left in its schedule. The network's decision is described in Appendix A.

Let T measure television exposure in the local market and $n(T)$ the share of the station's schedule occupied by the network bundle. The bundle includes programming and the advertisements sold by the network. As television becomes available, the network may reduce the bundle it supplies and the payments it makes to the station. The station takes the network's decision as given and chooses how to use the remaining airtime.

The station's schedule. I normalize the broadcast day to one. The network bundle occupies n of the schedule, while s is the time devoted to the station's own programming and a is the advertising time it sells directly. The schedule satisfies

$$n + s + a = 1, \qquad a \leq \chi s, \tag{1}$$

where $\chi > 0$ limits how much advertising the station can sell alongside its own programming. The station needs programming to attract listeners, so it cannot fill its entire schedule with commercials. It may sell less than this limit if advertising demand is weak.

Advertising demand. Local advertisers and national or regional advertisers purchase time from the station. Let q_L and q_N be the amounts bought by the two groups, so $a = q_L + q_N$. Both groups purchase advertising time using the station's spot rate card. I denote the common spot rate by p. Their demands are

$$\begin{aligned} q_L &= b\left[(v_L - \tau_L T)(1-a) + \rho s - p\right], \\ q_N &= b\left[(v_N - \tau_N T)(1-a) - p\right], \end{aligned} \tag{2}$$

where $b > 0$ measures how demand responds to price. The term $1 - a$ is the part of the broadcast day not sold directly by the station. It includes the station's own programming and the network bundle. I use it to capture how the station's schedule affects advertising demand, not to measure the number of listeners. The model does not determine whether changing programs raises or lowers listenership.

The parameters v_L and v_N describe how much each group values radio advertising before television. The parameters τ_L and τ_N capture the decline in that value as television becomes available. I assume $\tau_N > \tau_L \geq 0$: national and regional advertisers have more reason to switch to television than local businesses do. The term ρs, with $\rho > 0$, gives station-controlled programming an additional value to local advertisers. Local news or music, for example, may help a business reach customers in its area. The station does not choose a particular programming format in this model.

The station's choice. After the network decides how much airtime it will use, the station divides the rest between its own programs and advertising. Network payments are fixed from the station's point of view, so it chooses this allocation to maximize revenue from advertisements it sells directly. More commercials leave less time for other broadcasts, which affects what advertisers are willing to pay. I leave programming costs out of this choice.

I focus on the case in which the station would like to sell more advertising than its programming schedule allows, both before and after television entry. The advertising limit then binds, so $a = \chi s$. Combining this with the schedule gives

$$s^* = \frac{1-n}{1+\chi}, \qquad a^* = \frac{\chi(1-n)}{1+\chi}. \tag{3}$$

When the network withdraws sponsored programs, the station gains airtime and uses it for both its own programs and advertisements. This result depends on the advertising limit binding. If demand is too weak, the station need not sell more advertising.

Selling more commercials can put downward pressure on advertising rates, especially when television also gives advertisers another option. More station-controlled programming may partly offset that pressure by making radio more useful to local businesses. Under the conditions derived in the appendix, local advertising quantity increases, while the effect on national and regional advertising quantity is not determined. Local advertising revenue rises only if the additional sales make up for any decline in the rate. Nothing guarantees that these gains replace the network payments the station loses.

The FCC records report advertising revenue, not the number of advertisements sold, their prices, or the programs stations offered. The model therefore describes a possible explanation for the revenue changes rather than identifying these decisions separately. It also leaves out programming costs: a station could offer more of its own programming while spending less on it. Finally, this is a single-station model. It cannot explain how competing radio stations divide advertising sales, which I examine empirically through the market-structure comparisons.

4 Data

The analysis draws on three sets of data. The first is a newly constructed panel of commercial radio stations, with financial records that distinguish revenue from national networks, national and regional advertisers, and local advertisers. The second follows the expansion of television, from construction-permit applications filed before the FCC licensing freeze to the dates and locations of operating television stations. The third provides demographic and

economic characteristics of radio markets from the 1940 U.S. Census.

4.1 The American Radio Panel

The American Radio Panel is a longitudinal dataset of U.S. commercial radio stations developed jointly with Lena Song.[1] It combines station-level financial reports filed with the Federal Communications Commission (FCC) with information from contemporaneous editions of the *Broadcasting Yearbook*. The FCC reports provide revenues, expenses, and other financial outcomes for individual stations. The yearbooks identify the stations, their locations, and other characteristics needed to follow them over time.

Construction and coverage. The original FCC financial reports were professionally hand-transcribed from archival scans. I checked the transcribed records, reconciled inconsistencies in financial reporting across years, and linked them to the station listings in the *Broadcasting Yearbook*. The yearbooks record call signs, cities of license, counties, states, and other identifying information. I use these records to follow stations through changes in call signs, ownership, and frequency assignments, which would otherwise cause the same station to appear under different identities. Approximately 94 percent of station-year financial records can be linked to the longitudinal panel. Appendix Figure 12 reproduces an example of the original FCC reports.

The panel contains 4,324 commercial radio stations and 29,456 station-year observations before analysis-specific sample restrictions. Financial reports are available for 1946, 1947, 1948, 1950, 1952, 1954, 1955, 1956, 1957, 1958, and 1960. Coverage is irregular before 1955 and annual from 1954 through 1958, with the exception of 1959. I use the available reporting years without interpolating missing financial observations. All monetary variables are converted to constant 1950 U.S. dollars.

Revenue measures. The FCC accounts distinguish revenue from the sale of station time to networks from advertising sold directly by stations. Networks paid affiliates to carry commercial network programming, while advertisers could also purchase airtime on individual stations outside these arrangements. Network revenue is reported separately for the four major national networks, ABC, CBS, NBC, and MBS, regional networks, and other network arrangements. Non-network revenue is divided between advertising sold to national and regional customers and advertising sold to local customers.

[1]The American Radio Panel was developed jointly with Lena Song, Assistant Professor of Economics at the University of Illinois Urbana-Champaign.

I use four principal revenue outcomes. *Major-network revenue* consists of payments received from ABC, CBS, NBC, and MBS for station airtime. *National and regional advertising revenue* and *local advertising revenue* measure sales of station time to advertisers outside the network relationship, classified by the geographic scope of the customer. *Time-sales revenue* is the sum of these advertising revenues and payments from major, regional, and other networks. The categories are separately reported in the FCC accounts, allowing me to follow each source of revenue within a station.

Most network payments came from the four major networks. Among 25,175 station-year observations with complete network revenue information, 38.6 percent report positive major-network revenue, compared with 12.6 percent for regional networks and 7.1 percent for other networks. Average annual payments were approximately $11,600 from major networks, $826 from regional networks, and $471 from other networks. Major networks accounted for 89.9 percent of network revenue. Regional and other network payments together represented only 0.88 percent of total broadcast revenue. I therefore use major-network revenue to measure stations' financial dependence on the national networks, retaining the smaller network categories in total time-sales revenue.

Expenses and profitability. The FCC reports divide broadcast operating expenses into programming, selling, technical, and administrative costs. Talent expenses are reported separately within programming expenses and are not an additional component of total costs. I examine programming and talent expenses alongside selling expenses to measure changes in spending on broadcast content and advertising sales. The records report expenditures, but not the hours of programming produced, the quantity of advertising sold, or the prices paid for individual inputs.

I also use total broadcast revenue and total broadcast expenses to study changes in station scale and profitability. Total broadcast revenue includes receipts beyond the sale of station airtime and is therefore broader than time-sales revenue. Operating profit is the difference between total broadcast revenue and total broadcast expenses,

$$\text{Profit}_{it} = \text{Total Broadcast Revenue}_{it} - \text{Total Broadcast Expenses}_{it}, \tag{4}$$

where i denotes a radio station and t a reporting year.

4.2 Television Licensing and Signal Availability

I construct the television licensing records from the January 1949 edition of *Television Rates & Factbook: TV Directory No. 6*, published by the Radio News Bureau (Radio News Bureau,

1949). The directory lists proposed television stations and their regulatory status shortly after the FCC suspended application processing in September 1948. I digitized the listings using optical character recognition and manually checked the entries against the original directory. The resulting dataset contains 379 commercial television applications. Of these, 68 had received construction permits before September 30, 1948 and could proceed toward operation during the freeze. The remaining 311 applications were pending when processing stopped and could not receive authorization until licensing resumed.

To follow the subsequent operation of these stations, I use the television data constructed by Angelucci et al. (2024). The records provide transmitter locations, first broadcast dates, and Grade B signal contours for operating television stations. I link the January 1949 applications to these records using station identifiers and locations. The operating-station data also include stations absent from the freeze-era directory, allowing me to identify the first television signal reaching a radio market even when it came from a station outside the original application records.

I measure television availability using signal coverage. I geocode the cities of license of radio stations in the American Radio Panel and use each city centroid as the receiver location. A radio market receives television when its centroid falls within the Grade B contour of an operating television station. Among all stations whose signals reach the market, the earliest broadcast date determines the year of first television exposure. This measure captures television service available locally, including broadcasts transmitted from stations licensed in other cities.

I combine the geographic coverage and operating dates with the freeze-era application records to construct two treatment measures. The first is the year in which television initially became available in each radio market. The second identifies whether the television station responsible for that first exposure had received a construction permit before the freeze or had an application still pending in September 1948. I exclude markets already receiving television before the freeze and restrict the freeze-based research-design sample to markets whose first exposure can be linked to a classified freeze-era application. The benchmark comparison uses construction-permit status, while the staggered-adoption analysis uses the realized television-entry year. Appendix B.2 describes the matching procedure and treatment-assignment rules, including cases with missing call signs or overlapping television signals.

4.3 Market Characteristics

I use the 1940 U.S. Census to measure demographic and economic characteristics of radio markets before the expansion of commercial television. The county-level data, distributed

through the Inter-university Consortium for Political and Social Research (ICPSR), provide population, population density, urban and nonwhite population shares, educational attainment, and employment (Haines et al., 2010). I supplement these records with median family income and median years of schooling among adults aged 25 and older, constructed from the 1940 full-count Census microdata available through IPUMS USA (Ruggles et al., 2024). The data also include the share of adults aged 25 and older who completed high school and the total number of workers.

I match the Census characteristics to radio markets using county identifiers. The benchmark difference-in-differences specification includes four predetermined market characteristics: log population, urban population share, median family income, and median years of schooling. I allow their coefficients to vary by year, so that markets with different characteristics in 1940 can follow different outcome trends over the sample period.

4.4 Pre-Television Characteristics

Table 1 reports the distribution of radio-station financial outcomes in the research-design sample before the FCC freeze. I use observations from 1946–1948, covering 742 stations and 1,622 station-year observations with the revenue information required for the analysis. The number of observations differs across expense categories because some financial reports do not contain complete information for every outcome.

Radio stations varied widely in size. Mean annual time-sales revenue was approximately $173,000, compared with a median of $100,000. Local advertising was the largest source of revenue for the typical station. Mean local advertising revenue was about $88,000 and the median $74,000, compared with median national and regional advertising revenue of $9,900. Even stations at the 25th percentile of the local advertising distribution earned approximately $46,000 from local customers.

Network payments were distributed much less evenly. Mean major-network revenue was approximately $34,000, but the median was only $5,800 and the 25th percentile was zero. Stations at the 75th percentile received approximately $41,000 from the major networks. Thus, some stations entered the television period with a substantial source of revenue from the national networks, while others earned most of their revenue from advertising they sold themselves. The variation provides the basis for examining how television's effects differed with stations' initial network dependence.

Expenses and profit show similar dispersion. Mean total operating expenses were approximately $146,000, against a median of $92,000. At the median station, programming expenses were about $29,000, administrative expenses $34,000, selling expenses $11,000, and

technical expenses $16,000. Mean operating profit was approximately $40,000, nearly three times the median of $14,000. At the 25th percentile, profit was slightly negative. The typical station was substantially smaller than the mean station, and not all stations were profitable before television arrived.

5 Empirical Strategy

5.1 The FCC Freeze and the Benchmark Design

Television entered larger and more commercially attractive markets earlier, where radio stations also differed in size, advertising revenue, and dependence on national networks (Gentzkow, 2006; Koenig, 2023). These differences make the timing of television entry difficult to use on its own. Markets receiving television first may have followed different economic trends even without the new medium. I use the FCC licensing freeze of 1948–1952 to identify markets whose television entry was delayed by the interruption in federal authorization.

The freeze began on September 30, 1948, when the FCC suspended the processing of television-station applications. Some applicants had already received construction permits and could continue building their stations. Others were awaiting authorization and could not proceed until licensing resumed in 1952. Both groups had applied to enter television before the freeze, but the interruption prevented one group from moving forward while the other could begin broadcasting. This distinction provides the basis for comparing radio markets exposed to television at different times.

I construct the comparison using the regulatory status of the television station responsible for each market's first television signal. I exclude radio markets already receiving television before the freeze and identify the first operating television station whose Grade B signal reached each remaining market's city centroid. A market is classified as *early TV* if this station had received a construction permit before September 30, 1948, and as *freeze delayed* if its application was still pending. The classification is based on the regulatory status of the station that eventually provided television service, rather than on the presence of a television application within the radio market itself. Because I use subsequent station operation to identify the first signal, the comparison is not a pure intent-to-treat design based only on the applications filed before the freeze. Appendix B.2 describes the matching and classification rules.

The interruption produced a substantial difference in when television became available. Figure 4 shows the distribution of first television exposure across the two groups. Markets classified as early TV began receiving signals during the freeze as previously authorized

stations completed construction. Entry in freeze-delayed markets was concentrated after the FCC resumed licensing in 1952. The difference narrowed as television spread to the delayed markets. The design therefore compares earlier and later exposure to television, not markets that permanently received or never received the new medium.

I assign the market classification to commercial radio stations operating before the freeze. Appendix Table 12 reports the sample restrictions. The two groups were not identical before television arrived. Table 2 shows that early-TV markets had larger populations, higher urban shares, and higher incomes in 1940. These differences are present in the full freeze-design sample and among markets eligible for the incumbent-radio analysis. Construction permits were not randomly allocated, and the same characteristics that made markets attractive for early television entry may also have influenced radio revenues. I account for these predetermined differences in the regressions and examine whether the groups followed similar outcome trends before the freeze.

I begin with a difference-in-differences comparison based on the freeze-era regulatory classification. Let T_c equal one for early-TV markets and zero for freeze-delayed markets. I estimate

$$Y_{ict} = \alpha_i + \delta_t + \beta(T_c \times Post_t) + X_c'\Gamma_t + \varepsilon_{ict}, \tag{5}$$

where Y_{ict} is an outcome for radio station i in market c and reporting year t. The terms α_i and δ_t are station and reporting-year fixed effects, and $Post_t$ equals one for reporting years after 1948. The vector X_c contains four market characteristics measured in 1940: log population, urban population share, median family income, and median years of schooling. I interact these characteristics with reporting-year indicators, allowing their associations with radio outcomes to change over time. Standard errors are clustered at the city-state radio-market level.

The coefficient β measures the differential change in radio-station outcomes after 1948 between markets whose first television stations had received construction permits and those whose stations were delayed. It is a reduced-form comparison of the two regulatory groups. Television did not begin operating in every early-TV market at the same time, and many freeze-delayed markets eventually received television themselves. The estimate consequently does not measure the effect of a particular number of years of television exposure. Its causal interpretation requires that, absent the difference in television timing associated with the freeze, radio outcomes in the two groups would have followed parallel trends, conditional on the pre-television characteristics included in the specification.

I examine the evolution of this difference using a dynamic version of Equation 5, replacing the post-freeze interaction with interactions between the early-TV indicator and reporting-

year indicators. The omitted year is 1948, the last financial reporting year before the freeze. The pre-freeze coefficients show whether radio stations in the two groups were already changing differently, while the later coefficients trace their relative outcomes as television spread. The benchmark uses the regulatory interruption most directly, but it does not distinguish between stations that had just received television and those exposed for several years. For that, I turn to realized television-entry dates.

5.2 Staggered Television Exposure and Estimator Choice

Construction permits determined whether television stations could proceed during the freeze, but not when their signals first reached radio markets. Authorized stations still had to complete construction, and a television transmitter could serve several radio markets beyond its city of license. I therefore use the first year of local television signal availability to estimate how radio outcomes changed following entry. Let G_c denote the calendar year in which market c first received a Grade B television signal. Markets enter treatment in different years, producing a staggered-adoption design within the freeze-based research-design sample.

Treatment timing is measured in calendar years rather than FCC financial reporting periods. A market receiving television in 1953 is one year into treatment in its 1954 financial record, even though the panel contains no observation for 1953. For markets first receiving television late in 1952, I assign treatment to the 1953 cohort so that treatment corresponds to the first annual reporting period with meaningful exposure. Markets that remain without television during the estimation window contribute untreated observations to the comparison group.

A conventional two-way fixed-effects event study would compare radio outcomes across years relative to television entry. With staggered adoption, however, the resulting coefficients need not represent averages of the treatment effects of interest. Goodman-Bacon (2021) shows that the static TWFE estimator combines comparisons across treatment groups, including comparisons that use already-treated units as controls. de Chaisemartin and D'Haultfœuille (2020) shows that heterogeneous treatment effects can produce non-convex weights, while Sun and Abraham (2021) shows that dynamic event-study coefficients can reflect treatment effects from other relative periods. These concerns are relevant here because radio stations may respond differently to television depending on their initial network business, and the response may change with the time elapsed since entry.

Several alternative estimators allow treatment effects to differ across cohorts and exposure periods. Sun and Abraham (2021) estimate cohort-specific event-time effects and aggregate them across cohorts. Callaway and Sant'Anna (2021) construct group-time average treat-

ment effects, $ATT(g,t)$, using eligible comparison groups for each cohort and period. In my implementation of Callaway–Sant'Anna, markets that have not yet received television serve as controls. The number of these markets becomes smaller as television spreads. Combined with the irregular reporting years and the unbalanced station panel, this leaves some later cohort-time comparisons with relatively few untreated observations.

I use the imputation estimator of Borusyak et al. (2024) as the preferred staggered-adoption estimator and report Callaway–Sant'Anna estimates as a robustness check. Rather than estimating a separate comparison for every cohort and reporting year, BJS uses the available untreated station-years jointly to estimate the outcome process in the absence of television. It then predicts untreated outcomes for station-years observed after television entry. This makes use of the untreated observations across the panel rather than relying on a separate set of controls for each group-time effect. The estimator still requires sufficient untreated observations to identify the outcome process, and the validity of the predictions depends on how well that process describes what would have happened without television.

I specify untreated potential outcomes as

$$Y_{ict}(0) = \alpha_i + \delta_t + \varepsilon_{ict}, \tag{6}$$

where α_i and δ_t are station and reporting-year fixed effects. I estimate the fixed effects using observations from stations that have not yet received television. For each treated station-year, the fitted model gives an estimate of its outcome in the absence of television, $\widehat{Y}_{ict}(0)$. The imputed treatment effect is the difference between the observed outcome and this counterfactual,

$$\widehat{\tau}_{ict} = Y_{ict} - \widehat{Y}_{ict}(0). \tag{7}$$

I average these differences across treated station-years to obtain the overall treatment effect. I also aggregate them by years since television entry to examine the timing of the response. The event study reports estimates separately for event years 0 through 8 and pools subsequent observations into a 9+ category. Since stations enter treatment at different dates and the financial records are not available every year, the number and composition of stations contributing to each event-time estimate vary. The later estimates are based on fewer observations and should be interpreted with greater caution.

I examine pre-treatment outcomes using five placebo estimates for event years -5 through -1. These are deviations from the estimated untreated outcome process before a market receives television. Unlike the coefficients in a conventional event study, the BJS placebo estimates are not measured relative to a single omitted pre-treatment year. I report each

estimate and test whether the five coefficients are jointly equal to zero. Systematic differences before television entry would raise concerns about the counterfactual trends assumed by the estimator, although the absence of such differences cannot establish that the trends would have remained parallel after entry.

Standard errors are clustered at the city-state radio-market level. For the BJS estimates, I use the leave-one-cluster-out adjustment whenever the data permit its computation. I report financial outcomes in levels and after applying the inverse hyperbolic sine (IHS) transformation to monetary values expressed in thousands of dollars. The level specifications retain the dollar scale of the FCC accounts, allowing changes in network revenue, advertising revenue, and total time-sales revenue to be compared directly. The IHS transformation accommodates zeros and reduces the influence of large differences in station size. Its coefficients, however, depend on the units of the underlying outcome, particularly when the data contain zeros, and cannot always be interpreted as percentage changes (Aihounton and Henningsen, 2021; Chen and Roth, 2024).

The staggered estimates use the actual arrival of television rather than the regulatory status of the station that eventually provided it. This brings treatment closer to the date when listeners and advertisers could use the new medium, but realized entry may also reflect decisions made after the freeze and economic conditions in the markets being served. BJS addresses the problem of estimating effects when treatment begins at different dates and effects may be heterogeneous; it does not make those dates exogenous. A causal interpretation requires that the untreated outcome model provides a valid counterfactual for the treated stations and that radio outcomes do not respond in anticipation of television entry. I use the benchmark freeze comparison, the pre-treatment placebo estimates, and the Callaway–Sant'Anna results to assess the sensitivity of the findings to these choices.

6 Results and Discussion

6.1 Benchmark Evidence from the FCC Freeze

I begin with the difference-in-differences comparison between radio stations in early-TV and freeze-delayed markets. Markets whose first television provider had received a construction permit before the September 1948 freeze could proceed toward television operation, while markets awaiting authorization were delayed. Figure 5 plots the unadjusted revenue series for the two groups. Revenue levels differed before the freeze, consistent with the differences in market characteristics documented in Table 2. I therefore compare changes in revenue after 1948, controlling for predetermined market characteristics as described in Section 5.1.

Table 3 reports an increase of $7,526 in annual local advertising revenue among stations in early-TV markets relative to those in freeze-delayed markets. This amounts to 9.7 percent of the pre-treatment control mean and is statistically significant at the 5 percent level. The IHS estimate is also positive and significant at the 10 percent level. Total time-sales revenue declines by an estimated $204, with a standard error of $6,328. Major-network revenue falls by $2,348 and national and regional advertising by $4,100, although neither estimate is statistically significant.

Figure 6 shows how these differences developed. The pre-freeze coefficients provide little evidence of different trends between the two groups. Local advertising increases in early-TV markets relative to delayed markets after 1948, with larger estimates in later years. The gradual increase is consistent with radio stations changing their programming and advertising sales over time. Replacing network programs would have required stations to fill the available airtime with other content and find advertisers willing to purchase spots alongside it. The financial records do not identify these decisions separately.

The interpretation also changes at longer horizons. Freeze-delayed markets began receiving television after the FCC resumed licensing in 1952, narrowing the difference in exposure between the groups. I interpret the benchmark as evidence on the earlier path to television availability associated with the freeze. The staggered analysis uses the realized year of television entry to estimate the response to local exposure.

6.2 Television Entry and Radio Revenue

Table 4 reports the preferred BJS estimates, with treatment beginning when television first becomes available in a radio market. Television entry reduces annual major-network revenue by $8,634 per station, or 31.5 percent of its pre-TV mean. Local advertising increases by $10,770, or 12.5 percent. Both estimates are statistically significant at the 5 percent level. National and regional advertising falls by $2,236, although the estimate is imprecise.

The estimated effect on total time-sales revenue is smaller: a decline of $756, with a standard error of $7,777, against a pre-TV mean of $158,077. The approximate 95 percent confidence interval ranges from a loss of $16,000 to a gain of $14,500. I can rule out declines much larger than 10 percent of average pre-TV time-sales revenue, but smaller losses and increases remain possible.

The revenue components changed more than their combined total. The $10,770 increase in local advertising is similar in magnitude to the $8,634 decline in major-network payments. The estimates are consistent with a shift toward advertising sold directly by radio stations, although the small and imprecise effect on total time-sales revenue does not establish that

stations fully replaced their lost network business.

The distinction between network-sponsored programming and station-sold advertising helps explain this pattern. National networks supplied affiliates with programs and the advertisements sold alongside them. If a sponsored program moved to television, the radio affiliate could lose the associated payment and regain control of that airtime. The station could then fill its schedule with its own programming and sell advertising directly. The network's sponsors moved with the program, while the station's advertising business remained tied to the demand for radio airtime.

Making more airtime available did not guarantee additional revenue. Stations needed programs that attracted listeners, whether through recorded music, news, or other station-controlled content. Television also offered advertisers another way to reach consumers, while additional radio advertising inventory could put downward pressure on spot rates. Lower prices might encourage local businesses to purchase more advertising, allowing stations to increase revenue if the additional sales outweighed the reduction in price. This is consistent with the local advertising gains in Table 4, although the FCC financial records do not report advertising rates, quantities sold, or programming hours.

National and regional spot advertising could respond differently. These advertisements were sold directly by stations, but television also competed for advertisers seeking audiences across multiple markets. Additional station-controlled airtime could increase the number of spots sold, while weaker demand could reduce their price. The estimated decline is imprecise, leaving the net effect on national and regional spot revenue unresolved.

I place greater weight on the level estimates for major-network revenue because many station-years report zero payments from the major networks. These are genuine zeros. They identify stations without revenue from this source of network business, and the distinction is important for the heterogeneity analysis below. Although the IHS transformation accommodates zero values, its interpretation is difficult when an outcome has a large mass at zero. The coefficient combines changes in whether stations receive network payments with changes in the amounts received, and its magnitude is sensitive to the units of measurement (Aihounton and Henningsen, 2021; Chen and Roth, 2024). The major-network IHS estimate is negative but imprecise, despite the statistically significant decline in levels. Local advertising has fewer zeros, and its IHS coefficient of 0.235 is positive and significant at the 5 percent level.

Figure 7 reports the dynamic estimates. Major-network revenue declines more at longer exposure horizons, while the local advertising estimates become more positive. The joint pre-treatment placebo tests do not reject for either outcome in levels. The gradual response is consistent with stations changing their network arrangements, programming, and adver-

tising sales over several years. The estimates do not establish the timing of these individual decisions.

Appendix Table 13 and Figures 19–20 report estimates using Callaway–Sant'Anna. The post-treatment estimates also show lower major-network revenue and higher local advertising. The joint pretrend test for major-network revenue in levels rejects under this estimator ($p = 0.008$). This raises concerns about the identifying assumptions for that outcome under the alternative specification, although the post-treatment estimates have the same direction.

6.3 Initial Network Dependence

The average revenue estimates combine stations with different relationships to the major networks. Some received payments from ABC, CBS, NBC, and MBS for carrying their programs. Others earned no revenue from these networks before television arrived. Stations receiving network payments had an existing source of business that could be lost as television became more attractive to the networks. The departure of sponsored programs could also leave them with additional airtime to sell to advertisers.

I measure initial network dependence using major-network revenue as a share of total time-sales revenue over 1946–1948. For station i, let $\mathcal{B}_i$ denote the available baseline observations. The baseline share is

$$s_i^N = \frac{\sum_{t \in \mathcal{B}_i} R_{it}^N}{\sum_{t \in \mathcal{B}_i} R_{it}^{TS}}, \tag{8}$$

where R_{it}^N denotes major-network revenue and R_{it}^{TS} denotes total time-sales revenue. I first divide stations into those with zero and positive baseline major-network shares.

These groups also differed in size before television arrived. In the aggregate-outcome sample, stations receiving major-network payments earned approximately $215,800 in average pre-TV total revenue, compared with $76,500 among stations receiving none. Positive-share stations also had higher operating expenses. Differences in their responses to television may therefore reflect initial revenue composition as well as other characteristics associated with station size.

Table 5 reports the estimates for the two groups, with Figure 9 showing the corresponding effects on major-network revenue and local advertising. Among zero-share stations, the estimated effects on these outcomes are $152 and $1,091, respectively. Neither is statistically significant. The local advertising estimate has a standard error of $3,418, with an approximate 95 percent confidence interval ranging from a decline of $5,600 to an increase of $7,800. National and regional spot advertising, however, increases by $5,152, significant at the 5 percent level.

Stations that initially received major-network payments experienced larger changes. Television entry reduces their annual major-network revenue by \$12,666, or 30.5 percent of its pre-TV mean. Local advertising increases by \$21,379, or 21.6 percent. Both estimates are statistically significant. The estimated gain in local advertising exceeds the decline in major-network payments, but total time-sales revenue increases by only \$1,927, with a standard error of \$9,966. The net effect on time-sales revenue is therefore imprecisely estimated.

The differences between the two groups are statistically significant for major-network revenue ($p = 0.023$), national and regional spot advertising ($p = 0.039$), and local advertising ($p < 0.001$). In particular, the estimated increase in local advertising among positive-share stations exceeds that among zero-share stations by \$20,288 annually. The pooled decline in major-network payments and increase in local advertising are concentrated among stations that had received network payments before television entered. Zero-share stations experienced little estimated change in local advertising, although their national and regional spot revenue increased.

This pattern is consistent with the airtime reallocation mechanism in the conceptual framework. Stations carrying sponsored network programs could use the time released by those programs to expand advertising sold directly to customers. Stations without major-network payments had less exposure to this particular source of disruption, although they could still respond to changes in radio advertising demand and prices. Their increase in national and regional spot revenue suggests that television affected stations through channels beyond the withdrawal of compensated major-network programming.

Figure 8 reports the dynamic estimates. Among positive-share stations, major-network revenue declines more at longer exposure horizons, while local advertising gains increase. The joint pre-treatment placebo tests do not reject in levels for major-network revenue ($p = 0.247$) or local advertising ($p = 0.836$). Local advertising among zero-share stations shows little estimated response across exposure horizons, and its pre-treatment placebo test does not reject ($p = 0.633$).

The pre-treatment evidence is less satisfactory for major-network revenue among zero-share stations. The joint placebo test rejects in levels ($p = 0.007$), making the small estimated effect for this group difficult to interpret causally. The corresponding test does not reject for positive-share stations, where the estimated decline in network revenue is concentrated. For local advertising, the differences between groups are statistically significant, and neither group's level pretrend test rejects. These comparisons provide stronger evidence of heterogeneous local advertising responses than the estimated change in major-network payments among zero-share stations.

6.4 Network Dependence and Local Advertising

Stations receiving major-network payments differed in how much of their revenue came from those networks. For some, payments from ABC, CBS, NBC, and MBS accounted for less than 5 percent of time-sales revenue; for others, they exceeded a quarter. Greater initial dependence meant that a larger share of the station's business was tied to the networks, although network revenue measures payments received, not the hours occupied by sponsored programming. I divide positive-share stations into quartiles of baseline network dependence, using cutoffs of 5.1, 13.6, and 24.6 percent.

Panel A of Table 6 reports the estimated effects on local advertising revenue across the four quartiles. Television entry increases annual local advertising revenue by \$30,098 in the lowest-dependence quartile and \$15,438 in the highest. The corresponding gains in the middle two quartiles are \$19,099 and \$21,333. All four estimates are statistically significant at the 1 percent level. The difference of \$14,660 between the lowest and highest quartiles is statistically significant ($p = 0.015$), although the joint test does not reject equality across all four groups at the 5 percent level ($p = 0.101$). Stations with the greatest initial reliance on major-network payments experienced smaller local advertising gains than those with the least reliance. The estimates do not, however, decline steadily across quartiles.

These differences are consistent with less network-dependent stations being better positioned to expand local advertising. Local advertising was already an important source of revenue across all four quartiles, but stations with smaller network businesses may have had programming and advertiser relationships more closely tied to local demand. As the conceptual framework suggests, lower radio advertising rates could encourage advertisers to purchase more spots, provided the additional sales compensated for the lower price. Stations with established local customers may have found it easier to take advantage of these opportunities. Those more dependent on network payments also had local advertising businesses, but expanding them further may have required greater changes to their programming and sales operations.

The opportunities to expand local advertising may also have varied across markets. A station seeking additional local advertisers faced different competitive conditions depending on whether its rivals relied on national networks or already operated without network payments. To examine these differences, I classify radio markets using the network status of incumbent stations in 1946–1948. Network-monopoly markets contain one station with positive major-network revenue and no observed zero-share incumbent. Mixed markets contain both positive- and zero-share stations. All-network competitive markets contain at least two positive-share stations and no observed zero-share incumbent.

Panel B of Table 6 and Figure 10 compare local advertising responses across these market

types within each dependence quartile. In Q1, television entry increases annual local advertising revenue by $52,753 among stations in all-network competitive markets, compared with $13,244 in mixed markets. The difference is statistically significant ($p = 0.011$). A similar pattern appears in Q2, where the estimated gains are $25,083 and $7,983, respectively ($p = 0.047$). The estimates remain larger in all-network competitive markets in Q3 and Q4, although the differences are imprecisely estimated. These pairwise tests are not adjusted for multiple comparisons.

Network-monopoly stations generally fall between these two groups, although their responses also vary across quartiles. Only five network-monopoly stations belong to Q4, limiting the precision of that comparison. Across all four quartiles, stations in mixed markets have smaller estimated local advertising gains than those in all-network competitive markets. Even among stations with similar initial network dependence, the estimated local advertising response varied with the types of radio competitors already operating in the market.

Stations without major-network payments may already have served local advertisers that network-connected stations hoped to attract as they expanded their own advertising sales. This could make additional local revenue harder to earn in mixed markets. The comparisons also reflect differences between markets, including population and other characteristics associated with local advertising demand. I examine these differences more closely in the following section.

6.5 Radio-Market Structure and Local Advertising

The quartile comparisons suggest that network-connected stations experienced smaller local advertising gains in mixed markets than in markets where all incumbent competitors received major-network payments. I now pool stations across dependence quartiles to examine these differences more closely and assess whether they are associated with market size and initial network dependence.

Table 7 reports the estimates, with Figure 11 plotting the corresponding effects. Without accounting for population, television entry increases annual local advertising revenue by $22,452 among network-connected stations in network-monopoly markets. The estimated gain is $10,528 in mixed markets and $31,121 in all-network competitive markets. The difference between mixed and all-network competitive markets is statistically significant ($p = 0.002$). Since both market types contain multiple incumbent stations, this pattern extends beyond a comparison between monopoly and competitive radio markets.

Market size may account for some of these differences. Larger markets could support more radio stations and provide a larger customer base for local advertising. They may

also have been more attractive to the major networks before television arrived. I therefore allow television's treatment effect to vary with log county population in 1940, alongside market structure. Population is centered at its sample mean and enters the treatment-effect projection; the untreated-outcome model continues to include station and year fixed effects.

Column (3) reports the population-adjusted estimates in levels. At the centering value of population, television increases annual local advertising revenue by \$28,873 in network-monopoly markets and \$28,827 in all-network competitive markets. The corresponding estimate for mixed markets is \$6,574, with a standard error of \$5,458. Although the mixed-market estimate is imprecise, its difference from the all-network competitive market effect remains statistically significant ($p = 0.002$). The joint test also rejects equality across market structures ($p = 0.004$). The IHS specification yields a similar difference between mixed and all-network competitive markets ($p = 0.019$).

Population is positively associated with the estimated local advertising response in levels. A one-unit increase in log 1940 county population is associated with an additional \$8,944 in the treatment effect, with a standard error of \$4,522. The corresponding IHS coefficient is 0.008 and is imprecisely estimated. The pooled pre-treatment placebo test does not reject in levels ($p = 0.836$), while the IHS test yields a p-value of 0.066.

Initial network dependence is another potential source of variation. Mixed markets contain a greater proportion of stations in the higher dependence quartiles than network-monopoly markets, as shown in Table 6. I therefore estimate an additional specification allowing treatment effects to vary additively with market structure, centered log 1940 population, and baseline network-dependence quartiles. This comparison uses the same 453 stations as the quartile analysis.

The estimated market-structure difference changes little. After accounting for population and network-dependence quartiles, the local advertising gain is \$21,963 smaller in mixed markets than in all-network competitive markets ($p = 0.003$). The corresponding difference in the population-only specification, estimated on the same sample, is \$22,252 ($p = 0.002$). The IHS comparison also remains statistically significant ($p = 0.017$). Joint tests reject equality of the market-structure coefficients in levels ($p = 0.008$) and IHS ($p = 0.043$), while the quartile coefficients are not jointly significant in either specification. The smaller local advertising response in mixed markets thus persists after accounting for these observed differences in population and initial network dependence.

These estimates are consistent with the possibility that stations operating without major-network payments had already established local advertising businesses that their network-connected competitors found difficult to expand into. Yet market structure was not randomly assigned, and other characteristics of mixed markets could also be associated with the re-

sponse to television. The estimates do not distinguish competition for existing advertisers from differences in programming or advertising rates.

I also compare network-connected and zero-share stations operating within the same mixed markets. Table 8 includes station and market-by-year fixed effects, comparing changes in local advertising revenue between the two types of stations within a market-year. Following television entry, network-connected stations experience a $20,467 smaller change in local advertising revenue than their zero-share competitors, significant at the 1 percent level. Unlike the preceding BJS estimates, this specification measures the relative response of the two station types and does not identify the absolute treatment effect for either group.

The difference varies with the number of incumbent stations. In mixed markets with two stations, the estimated relative change is $-\$5,087$, with a standard error of $7,977. In markets with more than two stations, network-connected stations experience a $27,517 smaller change in local advertising revenue, significant at the 1 percent level. The difference between these estimates is statistically significant ($p = 0.041$). The relative disadvantage of network-connected stations is therefore more pronounced in mixed markets containing more incumbents.

The station-level results leave open whether television increased the total local advertising revenue earned by radio stations within a market. Table 9 reports an estimated increase of $2,350 per market-year, with a standard error of $7,266. The estimates for incumbent and entrant stations are also imprecise. I cannot determine whether television expanded the local radio advertising market or redistributed advertising revenue among stations. I also find no statistically significant change in the number of observed radio stations following television entry. Appendix Table 15 reports an estimated decline of 0.076 stations per market under BJS, with a standard error of 0.223. The estimates do not rule out changes in station entry or exit.

6.6 Operating Expenditures

The increase in station-sold advertising may have been accompanied by changes in programming and sales expenditures. Stations losing network-sponsored programs could fill the available airtime with their own content, while selling more advertising directly might require additional sales effort. I examine these responses using the FCC financial records, which report programming, talent, and selling expenses. Table 10 presents the estimates separately for stations with zero and positive baseline major-network revenue.

Among stations that receive major-network payments before television arrived, annual programming expenses decline by $9,372 and talent expenses by $3,910. Both level estimates

are statistically significant at the 1 percent level. The corresponding IHS estimates are also negative, significant at the 10 percent level for programming expenses and the 5 percent level for talent expenses. Stations with no baseline major-network revenue experience no statistically significant decline in either category. Their programming-expense estimate is positive but imprecise, while the talent-expense estimate is close to zero. The differences between the two groups are statistically significant for programming expenses ($p = 0.046$) and talent expenses ($p = 0.010$).

Selling expenses move in the opposite direction. Among positive-share stations, television entry increases annual selling expenses by $3,950, significant at the 10 percent level. The IHS estimate is also positive and statistically significant at the 5 percent level. Selling expenses increase among zero-share stations as well, although the level estimate is imprecise. The difference between the two groups is not statistically significant ($p = 0.738$).

The decline in programming and talent expenses is consistent with network-connected stations turning toward less costly programming. Recorded music, for example, could fill broadcast hours without the expense of producing live programs. Stations might also have reduced spending on talent as television competed for audiences. At the same time, higher selling expenses are consistent with greater effort to attract advertisers purchasing time directly from stations. These changes fit the shift toward local advertising documented earlier, although the expenditure records cannot establish how much additional airtime stations filled or what programs they offered. Lower programming expenses could reflect cheaper content, fewer programming hours, or reductions in other production costs.

6.7 Aggregate Financial Outcomes

The changes in individual revenue and expenditure categories leave open whether stations were financially better off after television entered. I examine total broadcast revenue, operating expenses, and profit, defined as the difference between total revenue and expenses. Total broadcast revenue includes receipts beyond station time-sales revenue.

Table 11 reports an estimated increase of $6,072 in annual total revenue, with a standard error of $16,578. Total expenses increase by an estimated $3,985, with a standard error of $10,875, while profit increases by $2,086, with a standard error of $6,579. None of these level estimates is statistically significant, and the joint pre-treatment placebo tests do not reject for any of the three outcomes.

The estimates are too imprecise to determine whether television improved or worsened overall station finances. The approximate 95 percent confidence interval for total revenue ranges from a decline of $26,400 to an increase of $38,600. For profit, it ranges from a loss

of $10,800 to a gain of $15,000. Changes in the composition of revenue and spending may have offset one another, but the evidence does not establish whether stations preserved their profitability.

I place a higher weight on the level estimates because profit can take negative values and its accounting relationship with revenue and expenses is defined in dollars. The IHS profit estimate is positive and statistically significant, despite the imprecise level estimate. With zero and negative observations, however, the transformed coefficient does not have a straightforward percentage interpretation. The IHS specification for total revenue also rejects the joint pre-treatment placebo test ($p = 0.016$), warranting caution in interpreting that estimate.

7 Conclusion

This paper studies how commercial radio stations responded to the arrival of television. Radio stations earned revenue from national networks, local advertisers, and national and regional advertisers purchasing airtime directly from individual stations. Television affected these sources of revenue differently. Using the staggered introduction of television following the FCC licensing freeze and station-level financial records, I examine how radio stations adjusted their revenue sources and operating expenditures as television entered local markets.

I find that television entry reduced annual major-network revenue by $8,634 per station, or 31.5 percent of its pre-TV mean, while local advertising increased by $10,770, or 12.5 percent. The estimated change in total time-sales revenue is a decline of only $756. However, this estimate is imprecise, with a 95 percent confidence interval ranging from a decline of approximately $16,000 to an increase of $14,500. The evidence is therefore stronger for changes in the sources of radio revenue than for their combined effect.

The institutional relationship between networks and their affiliates provides one explanation for these changes. National networks supplied sponsored programming and paid radio stations to carry it. When these programs moved to television, stations could lose network payments while gaining control of the airtime. They could then use this time for their own programming and sell advertising alongside it. Television may also have reduced radio advertising rates by giving advertisers another medium through which to reach consumers. Lower rates could encourage local businesses to purchase more advertising, but stations still needed programming that attracted listeners those businesses wanted to reach. Additional programming and lower advertising rates could therefore work together to increase local advertising sales. The financial records do not measure programming hours, advertising quantities, or prices, so I cannot determine how much each contributed to the observed revenue increase.

The response differed considerably across stations. The decline in network payments and the increase in local advertising were concentrated among stations that had received major-network revenue before television arrived. Within this group, stations in the lowest quartile of initial network dependence gained approximately $30,100 in annual local advertising, compared with $15,400 in the highest quartile. This difference is statistically significant, although local advertising gains did not decline steadily across quartiles. Greater initial reliance on network payments therefore did not translate into larger gains in local advertising. The opportunities to expand this business also depended on the markets in which stations operated.

Local advertising responses also differed depending on which radio stations were already operating in a market. After accounting for 1940 population, the estimated increase in local advertising was $6,600 for network-connected stations in mixed markets, compared with $28,800 in all-network competitive markets. This difference changes little when I also account for stations' initial network dependence. One possible explanation is that stations without major-network payments had already built relationships with local advertisers and developed programming suited to local audiences. Their network-connected competitors may therefore have found it harder to attract those advertisers when television arrived. The data do not show whether these changes in programming or advertiser relationships occurred, and other differences between markets may also explain the results.

Changes in operating expenditures provide further evidence that stations adjusted how they operated. Among network-connected stations, annual programming expenses fell by $9,372 and talent expenses by $3,910, while selling expenses increased by $3,950. These results are consistent with stations adopting less expensive programming and devoting more resources to selling advertising directly. They do not establish which programs stations offered or whether programming hours increased. The estimated effects on total broadcast revenue, operating expenses, and profit are also imprecise, leaving the overall consequences for station finances uncertain.

The radio–television transition illustrates how technological change can affect different activities within the same firm. Losing an established source of revenue does not necessarily eliminate the firm's other opportunities to earn revenue. Radio stations could continue selling advertising even as television displaced part of their network business. But developing that business required access to listeners and advertisers, and the opportunities differed across stations and markets. The findings show that television changed where radio stations earned their revenue, but the estimates do not establish whether these changes left stations financially better or worse off.

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Tables and Figures

Tables

Table 1: Radio Station Finances, 1946–1948

	Mean	Std. Dev.	P25	Median	P75	Observations
Panel A: Revenue (1950 $)						
Total revenue	186,587	254,005	59,471	104,786	207,350	1,622
Time-sales revenue	173,267	217,433	56,575	100,313	202,560	1,622
Major-network revenue	34,240	64,148	0	5,805	40,929	1,622
National/regional advertising revenue	48,764	118,197	1,011	9,931	36,206	1,622
Local advertising revenue	88,079	65,527	46,374	73,625	113,801	1,622
Panel B: Expenses (1950 $)						
Total expenses	145,991	176,540	56,397	91,582	165,382	1,609
Programming expenses	48,523	68,227	17,981	28,675	51,841	1,605
Talent expenses	7,466	22,850	69	1,386	5,613	1,603
Selling expenses	20,515	27,987	5,812	11,462	24,702	1,605
Administrative expenses	51,892	63,193	18,719	34,403	61,828	1,606
Technical expenses	25,282	28,085	10,234	15,900	29,205	1,606
Panel C: Profitability (1950 $)						
Profit	40,434	99,250	-232	14,445	45,154	1,609

Notes: The table reports summary statistics for radio stations observed during 1946–1948, before the divergence in television exposure induced by the FCC freeze. The sample is drawn from the common incumbent-radio research-design population and is restricted to station-year observations with the revenue information required for the revenue analysis. All monetary variables are reported in constant 1950 U.S. dollars. Total revenue denotes total broadcast revenue reported by the station. Time-sales revenue is defined as the sum of major-, regional-, and other-network revenue and national/regional and local advertising revenue. Because only major-network revenue is reported separately in Panel A, the displayed revenue components do not sum exactly to time-sales revenue. Expense statistics use all nonmissing, nonnegative observations for each expense outcome within this sample, so the number of observations may differ across outcomes. Profit is defined as total revenue minus total expenses and is permitted to take negative values. The pre-treatment revenue sample contains 742 radio stations and 1,622 station-year observations.

Table 2: Balance in Predetermined Market Characteristics

	Freeze-delayed	Early TV	Difference	Std. diff.	p-value
Panel A: Full FCC Freeze Design Universe					
Log population	10.542	10.899	0.357	0.421	0.001
Urban population share	35.13	41.90	6.76	0.295	0.001
Median income	474.1	612.2	138.1	0.404	0.001
Median years of schooling	7.547	7.709	0.163	0.144	0.034
Number of markets	430	436			
Panel B: Incumbent-Radio Analysis Markets					
Log population	10.708	11.106	0.398	0.481	0.001
Urban population share	41.51	47.30	5.79	0.263	0.005
Median income	528.7	689.2	160.5	0.477	0.001
Median years of schooling	7.664	7.788	0.124	0.119	0.205
Number of markets	217	249			

Notes: The unit of observation is the radio market, and each market is counted once. Panel A uses the full set of 866 markets for which first television entry can be classified using the FCC freeze-era application records: 436 early-TV markets and 430 freeze-delayed markets. Panel B restricts the sample to the 466 markets eligible for the incumbent-radio analysis: 249 early-TV markets and 217 freeze-delayed markets. Panel B does not condition on subsequent revenue-data availability or final regression-sample restrictions. Urban population share is expressed on a 0–100 scale, and median income is reported in 1940 U.S. dollars. The difference column reports the early-TV mean minus the freeze-delayed mean. The standardized difference divides this mean difference by the square root of the average of the two group variances. The final column reports p-values from market-level regressions of each characteristic on the early-TV indicator using heteroskedasticity-robust standard errors.

Table 3: Early Television Exposure and Radio Revenue

Panel A: Levels (1950 $)				
	Time-sales revenue	Major-network revenue	National/regional advertising revenue	Local advertising revenue
	(1)	(2)	(3)	(4)
Early TV × Post-1948	-204	-2,348	-4,100	7,526**
	(6,328)	(2,532)	(3,232)	(3,142)
Pre-treatment control mean	130,949	23,909	28,160	77,195
Joint pretrend p-value	0.566	0.598	0.335	0.457
Panel B: IHS				
	Time-sales revenue	Major-network revenue	National/regional advertising revenue	Local advertising revenue
	(1)	(2)	(3)	(4)
Early TV × Post-1948	0.0045	0.0342	-0.1227	0.0954*
	(0.0494)	(0.0884)	(0.0809)	(0.0544)
Pre-treatment control mean	5.149	2.206	2.525	4.745
Joint pretrend p-value	0.639	0.694	0.137	0.414
Observations	7,192	7,192	7,192	7,192
Radio stations	826	826	826	826
Clusters	451	451	451	451
Radio station FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
1940 market controls × Year	✓	✓	✓	✓

Notes: This table reports difference-in-differences estimates comparing incumbent radio stations in early-TV markets with stations in freeze-delayed markets, which form the comparison group. Early TV × Post-1948 equals one for stations in early-TV markets after 1948. Panel A reports revenue outcomes in constant 1950 U.S. dollars. Time-sales revenue is defined as the sum of major-, regional-, and other-network revenue and national/regional and local advertising revenue; column (2) isolates revenue received from the major national networks. Panel B reports inverse hyperbolic sine transformations of the same outcomes after expressing revenues in thousands of constant 1950 U.S. dollars. The pre-treatment control mean is calculated for freeze-delayed markets during the pre-treatment period. The joint pretrend p-value tests whether the pre-treatment interaction coefficients are jointly equal to zero. All specifications include radio-station and year fixed effects and allow predetermined 1940 market characteristics—log population, urban share, median income, and median years of schooling—to vary by calendar year. Standard errors are clustered by radio market. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Television Entry and Radio Revenue

Panel A: Levels (1950 $)	Time-sales revenue	Major-network revenue	National/regional advertising revenue	Local advertising revenue
	(1)	(2)	(3)	(4)
Average treatment effect	-756	-8,634**	-2,236	10,770**
	(7,777)	(4,145)	(2,716)	(4,343)
Pre-TV mean of outcome	158,077	27,420	42,701	86,045
Joint pretrend p-value	0.932	0.615	0.362	0.976
Panel B: IHS	Time-sales revenue	Major-network revenue	National/regional advertising revenue	Local advertising revenue
	(1)	(2)	(3)	(4)
Average treatment effect	0.1163	-0.2053	-0.0062	0.2352**
	(0.0906)	(0.2083)	(0.1443)	(0.1001)
Pre-TV mean of outcome	5.274	2.142	2.826	4.836
Joint pretrend p-value	0.057	0.842	0.465	0.129
Observations	6,861	6,861	6,861	6,861
Treated observations	4,462	4,462	4,462	4,462
Untreated observations used	2,399	2,399	2,399	2,399
Radio stations	769	769	769	769
Clusters	444	444	444	444
Radio station FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: This table reports average treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator for the freeze-based research-design sample. Treatment begins in the calendar year in which a radio market first receives television exposure, and the reported ATT averages treatment effects over observed post-treatment station-years. Panel A reports revenue outcomes in constant 1950 U.S. dollars. Time-sales revenue includes payments from major, regional, and other networks as well as national/regional and local advertising revenue; column (2) isolates revenue received from the major national networks. Panel B reports inverse hyperbolic sine transformations of the same outcomes after expressing revenues in thousands of constant 1950 U.S. dollars. Pre-TV means are calculated over observations preceding each station's own television-entry date in the corresponding estimation sample; the Panel B means therefore refer to the transformed outcomes rather than transformations of the Panel A means. The untreated potential-outcome model includes radio-station and year fixed effects. The joint pretrend p-value tests whether the five BJS placebo coefficients are jointly equal to zero. Standard errors are clustered by radio market and use the leave-one-cluster-out adjustment. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effect of TV Entry by Baseline Major-Network Revenue Share.

	Time-sales revenue (1)	Major-network revenue (2)	National/regional advertising revenue (3)	Local advertising revenue (4)
Panel A: Levels (1950 $)				
Zero baseline major-network share	7,151	152	5,152**	1,091
	(4,470)	(272)	(2,172)	(3,418)
Positive baseline major-network share	1,927	-12,666**	-5,291	21,379***
	(9,966)	(5,645)	(4,398)	(3,682)
Positive vs. zero p-value	0.640	0.023	0.039	< 0.001
Joint pretrend p-value: Zero	0.516	0.007	0.559	0.633
Joint pretrend p-value: Positive	0.791	0.247	0.310	0.836
Panel B: IHS				
Zero baseline major-network share	0.0500	-0.0644	0.3850**	0.0606
	(0.0531)	(0.1160)	(0.1660)	(0.0660)
Positive baseline major-network share	0.3108**	-0.0299	0.0744	0.4959***
	(0.1250)	(0.2440)	(0.1440)	(0.1160)
Positive vs. zero p-value	0.055	0.898	0.162	0.001
Joint pretrend p-value: Zero	0.203	0.004	0.357	0.748
Joint pretrend p-value: Positive	0.582	0.720	0.771	0.066
Zero baseline-share sample	2,362 observations; 286 radio stations; 254 clusters			
Positive baseline-share sample	4,331 observations; 454 radio stations; 266 clusters			
Radio station FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: This table reports heterogeneous treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator for the freeze-based research-design sample. Stations are classified according to their baseline major-network revenue share over 1946–1948. The baseline share is constructed as the ratio of major-network revenue to time-sales revenue across usable baseline observations. Stations with a baseline share equal to zero comprise the zero-share group; stations with a strictly positive baseline share comprise the positive-share group. Treatment effects are estimated separately for the two groups. Panel A reports revenue outcomes in constant 1950 U.S. dollars. Time-sales revenue includes payments from major, regional, and other networks, as well as national/regional and local advertising revenue; column (2) isolates revenue received from the major national networks. Panel B reports inverse hyperbolic sine transformations of the same outcomes, after expressing revenues in thousands of constant 1950 U.S. dollars. The positive-versus-zero p-value tests equality of the subgroup-specific treatment effects using a joint BJS heterogeneity specification with radio-station fixed effects and baseline-group-specific year fixed effects. The subgroup pretrend p-values are joint tests of the five BJS placebo coefficients from the corresponding subgroup-specific estimations. Standard errors are clustered by radio market. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Heterogeneity in Local Advertising Effects by Baseline Network Dependence and Radio-Market Structure

	Baseline major-network revenue-share quartile			
	Q1 $< 5.1\%$	Q2 5.1–13.6%	Q3 13.6–24.6%	Q4 $> 24.6\%$
Panel A: By baseline network dependence				
Television entry	30,098*** (5,375)	19,099*** (3,905)	21,333*** (6,623)	15,438*** (4,975)
Equal effects across quartiles (p)		0.101		
Q1 = Q4 (p)		0.015		
Panel B: By baseline dependence and radio-market structure				
Network monopoly	23,966*** (4,283)	24,599*** (4,797)	12,456 (12,586)	16,253* (9,783)
Mixed market	13,244 (8,280)	7,983 (5,351)	12,516 (7,941)	9,748 (8,130)
All-network competition	52,753*** (13,657)	25,083*** (7,440)	32,228*** (11,194)	21,201*** (5,283)
Equal market-structure effects (p)	0.040	0.021	0.294	0.443
Sample	4,327 observations; 453 stations; 266 clusters			
Radio station FE	✓			
Year FE	✓			

Notes: The table reports Borusyak–Jaravel–Spiess (BJS) estimates of television-entry effects on annual local advertising revenue, in constant 1950 U.S. dollars. Baseline network dependence is measured as the ratio of major-network revenue to time-sales revenue over 1946–1948. The sample contains 453 incumbent radio stations with positive baseline major-network revenue shares, excluding one station with a 100% share. Quartiles are defined at the station level. Panel A reports treatment effects by baseline network-dependence quartile. Panel B further distinguishes stations by pre-TV radio-market structure. Network-monopoly markets contain one positive-share incumbent and no zero-share incumbent; mixed markets contain both types; all-network competitive markets contain at least two positive-share incumbents and no zero-share incumbent. Market classifications are constructed before excluding the 100%-share station. Each panel is estimated separately, but jointly across its respective groups, using station and year fixed effects in the BJS untreated-outcome model. Panel A reports four quartile-specific effects; Panel B reports twelve quartile-by-market-structure effects. In Panel A, the equality test compares effects across all four quartiles. In Panel B, equality tests compare the three market-structure effects within each quartile. Mixed and all-network competitive markets differ in Q1 ($p = 0.011$) and Q2 ($p = 0.047$). The pooled pretrend p-value is 0.843 in both panels. Q4 contains only five network-monopoly stations. Standard errors are clustered by radio market and reported in parentheses. Pairwise p-values are unadjusted for multiple testing. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Heterogeneity in Local Advertising Effects by Pre-TV Radio-Market Structure

	Market structure only		+ 1940 population	
	Levels (1950 $)	IHS	Levels (1950 $)	IHS
	(1)	(2)	(3)	(4)
TV-entry effect: network-monopoly markets	22,452***	0.610***	28,873***	0.616***
	(3,841)	(0.136)	(5,345)	(0.151)
Mixed market vs. network monopoly	−11,924**	−0.261***	−22,299***	−0.270*
	(5,104)	(0.094)	(7,175)	(0.146)
All-network competition vs. network monopoly	8,669	−0.064	−46	−0.072
	(5,839)	(0.098)	(5,226)	(0.132)
Log 1940 county population (ATT slope)			8,944**	0.008
			(4,522)	(0.075)
Mixed vs. all-network (p)	0.0022	0.0111	0.0024	0.0192
Joint market-structure test (p)	0.0065	0.0055	0.0044	0.0438
Pooled pretrend test (p)	0.8359	0.0661	0.8359	0.0661
Sample	4,331 observations; 454 radio stations; 266 clusters			
Radio station FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: The table reports Borusyak–Jaravel–Spiess (BJS) estimates of television-entry effects on local advertising revenue for incumbent radio stations with positive baseline major-network revenue shares (4,331 observations; 454 stations). Market structure is defined using 1946–1948 financial data. Network-monopoly markets contain one positive-share incumbent and no zero-share incumbent; mixed markets contain both positive- and zero-share incumbents; all-network competitive markets contain at least two positive-share incumbents and no zero-share incumbent. The first row reports the television-entry effect for stations in network-monopoly markets; subsequent coefficients report differences relative to this group. Columns (3)–(4) additionally allow treatment effects to vary with centered log 1940 county population, with estimates evaluated at the centering value. Population enters the treatment-effect projection, not the untreated-outcome model. Revenue levels are in constant 1950 U.S. dollars; IHS is asinh(revenue/1000). Reported tests compare mixed and all-network effects, jointly test market-structure differences, and assess pooled BJS pretrends. The untreated-outcome model includes station and year fixed effects. Standard errors are clustered by radio market and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Local Advertising Responses to Television Entry in Mixed Radio Markets

	Local Advertising Revenue	
	(1) Pooled	(2) By Number of Stations
Network-connected × Post-TV	$-20,467^{***}$	
	(5,729)	
Network-connected × Post-TV × (2 stations)		$-5,087$
		(7,977)
Network-connected × Post-TV × (>2 stations)		$-27,517^{***}$
		(7,221)
Joint pretrend p-value	0.127	
Differential pretrend p-value		0.603
Equality of post-TV effects (p)		0.041
Observations	2,132	2,132
Radio stations	250	250
Markets	76	76
Radio station FE	✓	✓
Market × Year FE	✓	✓

Notes: The dependent variable is station-level local advertising revenue in constant 1950 U.S. dollars. The sample is restricted to mixed radio markets, defined as markets containing both network-connected and zero-network stations before television entry. Network-connected stations are identified by positive baseline major-network revenue. Column (1) estimates the differential post-TV response of network-connected stations relative to zero-network stations within the same market-year. Column (2) allows this differential response to vary with the number of incumbent radio stations in the market before television entry. The sample includes 35 two-station markets (70 stations) and 41 markets with more than two stations (180 stations). Both specifications include station and market-by-year fixed effects. Standard errors are clustered by radio market and reported in parentheses. The equality test compares the post-TV differential responses in two-station markets and markets with more than two incumbent stations. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Television Entry and Market-Level Local Advertising Revenue

	Local Advertising Revenue		
	(1) Total	(2) Incumbents	(3) Entrants
TV exposure	2,350	-354	2,704
	(7,266)	(5,963)	(7,017)
p-value	0.746	0.953	0.700
Joint pretrend p-value	0.618	0.767	0.589
Observations	3,678	3,678	3,678
Market FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: The dependent variable is total local advertising revenue aggregated across radio stations within each market-year, expressed in constant 1950 U.S. dollars. Column (1) reports the effect on total market-level local advertising revenue. Columns (2) and (3) separately report revenue earned by incumbent and entrant radio stations, respectively. Total local advertising revenue equals the sum of incumbent and entrant revenue. All specifications include market and year fixed effects. Standard errors are clustered by radio market and reported in parentheses.

Table 10: Effect of TV on Radio Station Operating Expenses

	Programming expenses		Talent expenses		Selling expenses	
	Levels	IHS	Levels	IHS	Levels	IHS
	(1)	(2)	(3)	(4)	(5)	(6)
Zero baseline major-network share	10,538	0.0208	-2	0.0909	5,656	0.0495
	(9,422)	(0.0901)	(895)	(0.1823)	(4,699)	(0.1274)
Joint pretrend p-value	0.162	0.092	0.503	0.350	0.794	0.098
Sample	2,345 observations; 284 radio stations; 253 clusters					
Positive baseline major-network share	-9,372***	-0.1118*	-3,910***	-0.2490**	3,950*	0.4364**
	(3,337)	(0.0607)	(1,146)	(0.1111)	(2,076)	(0.1975)
Joint pretrend p-value	0.507	0.453	0.523	0.540	0.589	0.322
Sample	4,306 observations; 454 radio stations; 266 clusters					
Positive vs. zero p-value	0.046	0.221	0.010	0.118	0.738	0.098
Radio station FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓

Notes: This table reports heterogeneous treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator for the freeze-based research-design sample. Stations are classified according to their baseline major-network revenue share over 1946–1948. The baseline share is constructed as the ratio of major-network revenue to time-sales revenue across usable baseline observations. Stations with a baseline share equal to zero comprise the zero baseline major-network-share group; stations with a strictly positive baseline share comprise the positive baseline major-network-share group. Treatment effects are estimated separately for the two groups. Columns (1), (3), and (5) report expenses in constant 1950 U.S. dollars; columns (2), (4), and (6) report inverse hyperbolic sine transformations after expressing expenses in thousands of constant 1950 U.S. dollars. Talent expenses are reported separately as a component of programming expenses. Sample counts are reported once for each baseline-share group because the estimation sample is the same across the six specifications within each group. The positive-versus-zero p-value tests equality of the subgroup-specific treatment effects using a joint BJS heterogeneity specification. The subgroup pretrend p-values are joint tests of the five BJS placebo coefficients from the corresponding subgroup-specific estimations. All subgroup specifications include radio-station and year fixed effects. Standard errors are clustered by radio market. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

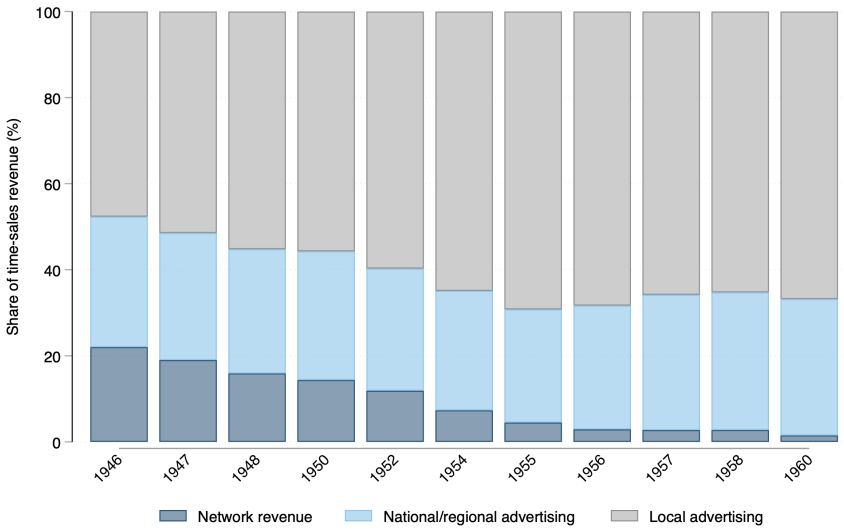
Table 11: Effect of TV on Total Revenue, Expenses, and Profits

	Total revenue		Total expenses		Profit	
	Levels	IHS	Levels	IHS	Levels	IHS
	(1)	(2)	(3)	(4)	(5)	(6)
Effect of TV entry	6,072	0.1091	3,985	0.0343	2,086	0.8930**
	(16,578)	(0.0925)	(10,875)	(0.0588)	(6,579)	(0.3743)
Joint pretrend p-value	0.233	0.016	0.275	0.058	0.154	0.061
Sample	6,833 observations; 767 radio stations; 444 clusters					
Radio station FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓

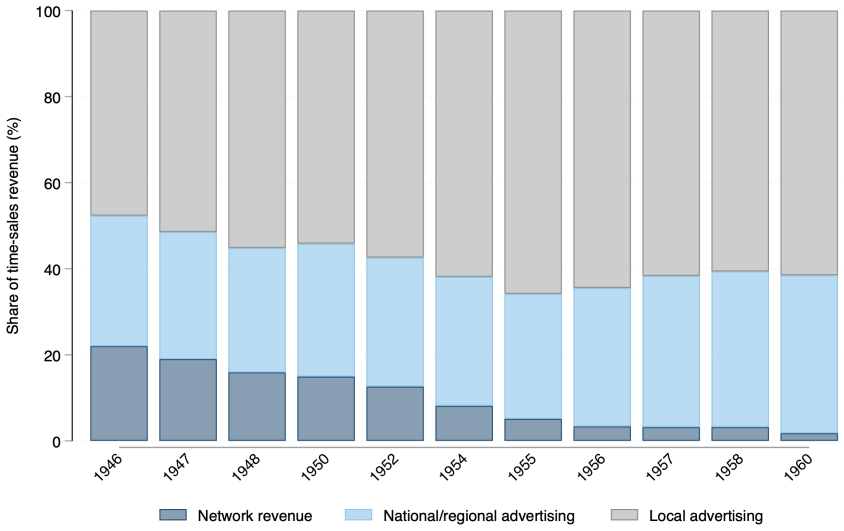
Notes: This table reports treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator for the freeze-based research-design sample. Columns (1), (3), and (5) report total revenue, total expenses, and profit in constant 1950 U.S. dollars; columns (2), (4), and (6) report inverse hyperbolic sine transformations of the same outcomes after expressing monetary values in thousands of constant 1950 U.S. dollars. Profit is defined as total broadcast revenue minus total broadcast expenses and may take negative values, so its IHS coefficient should not be interpreted as a conventional percentage effect. All columns use the same estimation sample. The untreated potential-outcome model includes radio-station and year fixed effects. The joint pretrend p-values test that the five BJS placebo coefficients are jointly equal to zero. Standard errors are clustered by radio market and use leave-one-cluster-out inference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figures

Figure 1: The Changing Revenue Structure of Commercial Radio



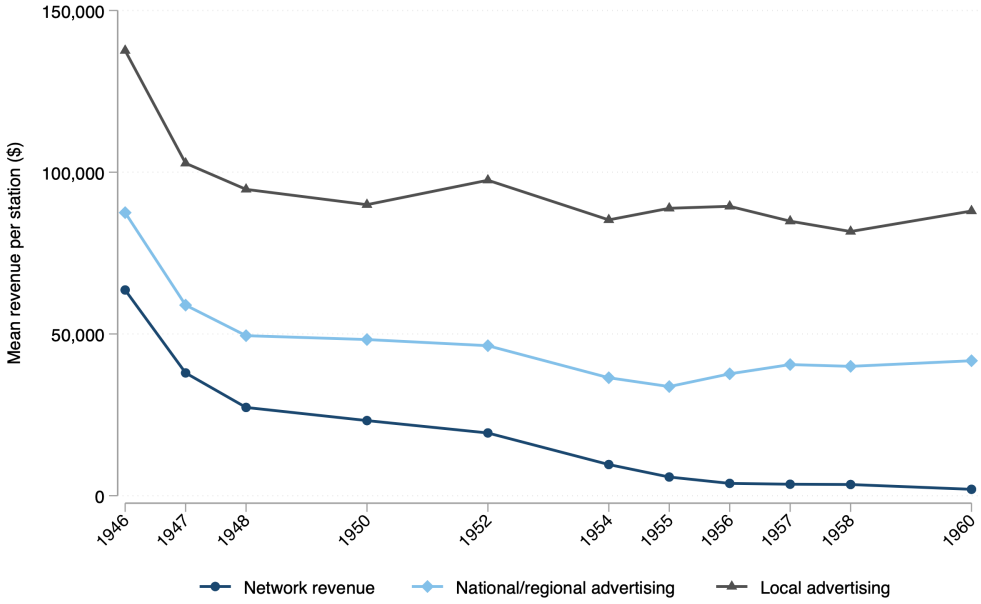
(a) All Radio Stations



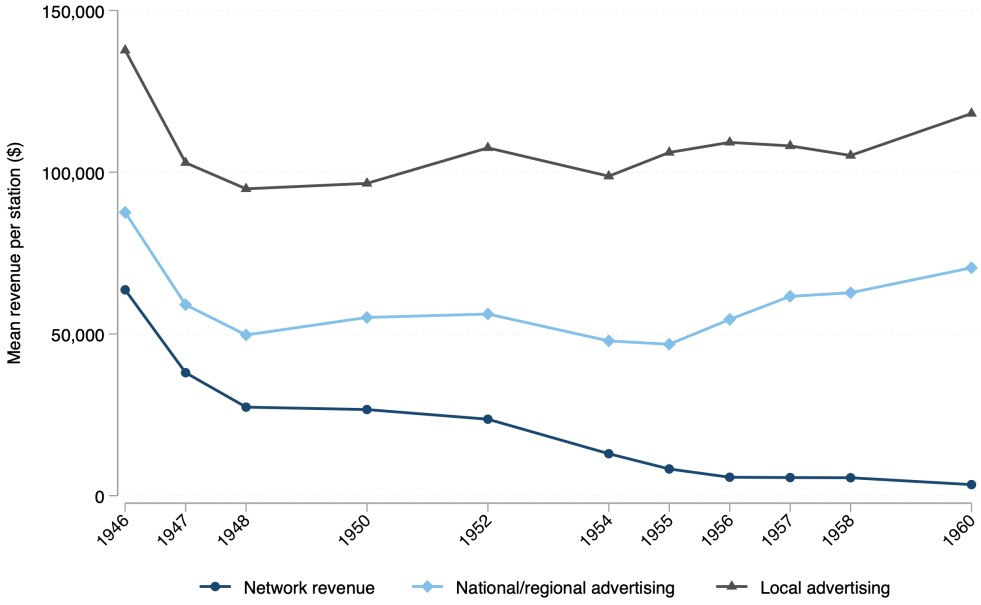
(b) Incumbent Radio Stations

Notes: Bars show the composition of aggregate radio time-sales revenue in available reporting years from 1946–1960. Panel A includes all observed stations; Panel B restricts to incumbent stations founded before 1949. Time-sales revenue is divided into network revenue, national/regional advertising, and local advertising, which sum to 100 percent. Network revenue includes major, regional, and other network arrangements.

Figure 2: The Changing Revenue Structure of Commercial Radio: Revenue Levels



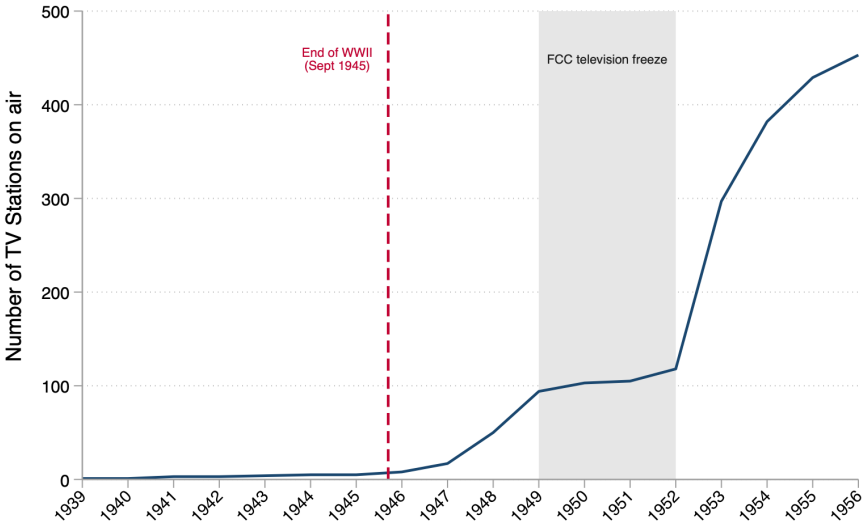
(a) All Radio Stations



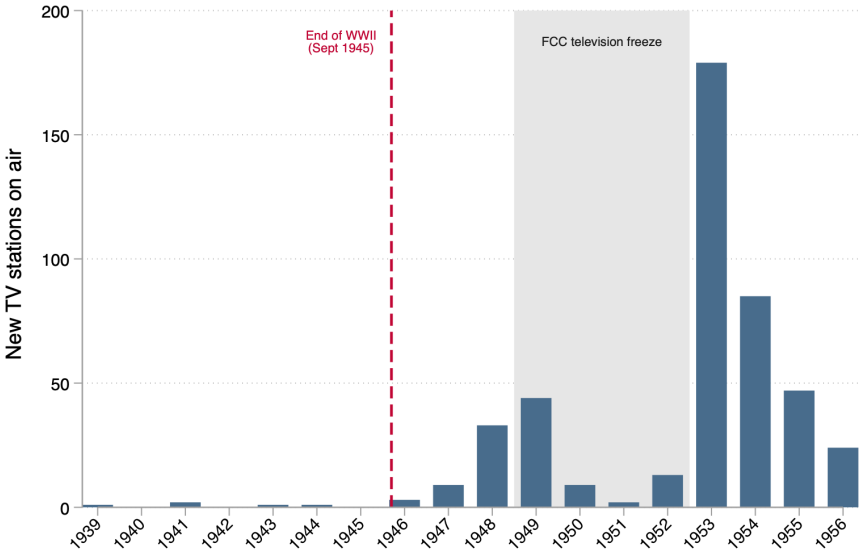
(b) Incumbent Radio Stations

Notes: Lines show mean revenue per station by source in available reporting years from 1946–1960. Panel A includes all observed stations; Panel B restricts to incumbent stations founded before 1949. Network revenue includes major, regional, and other network arrangements. Revenues are reported in constant 1950 U.S. dollars.

Figure 3: Television Expansion and the FCC Freeze



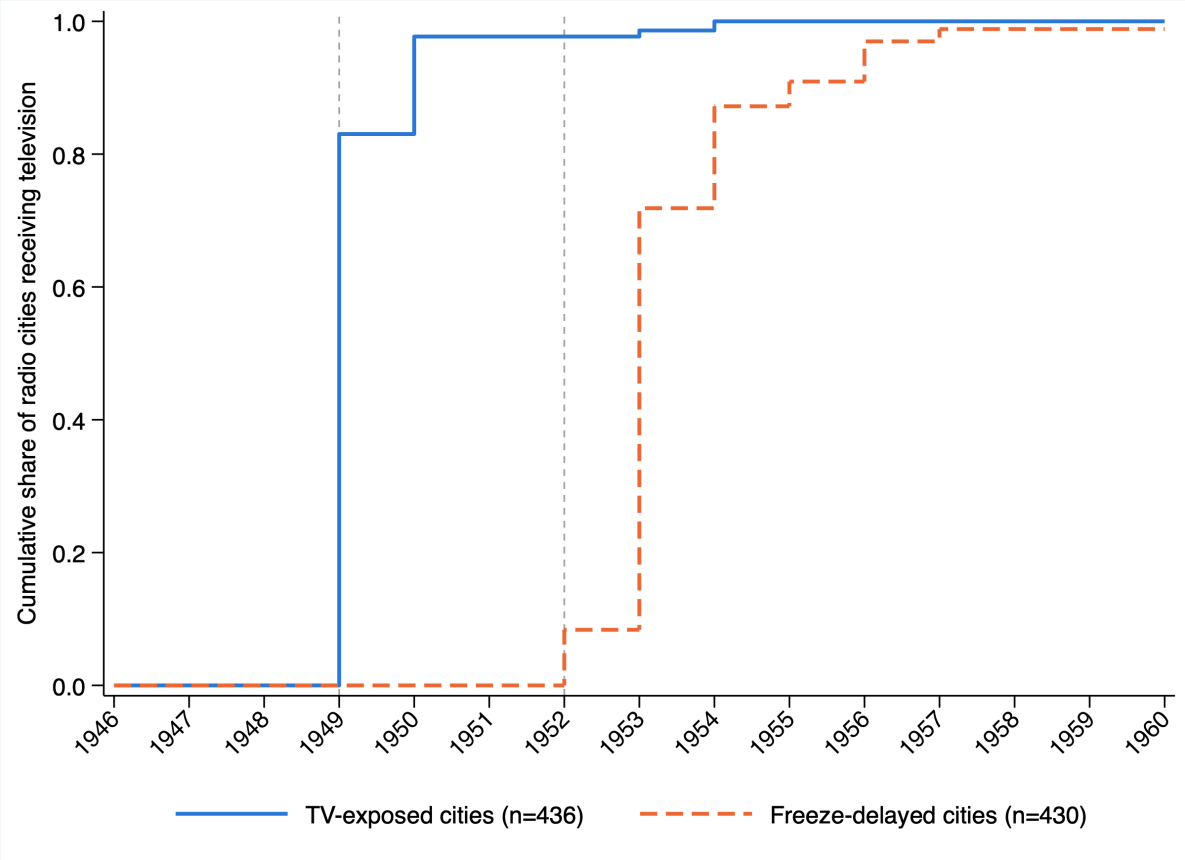
(a) Cumulative Television Stations on Air



(b) New Television Stations Beginning Operation

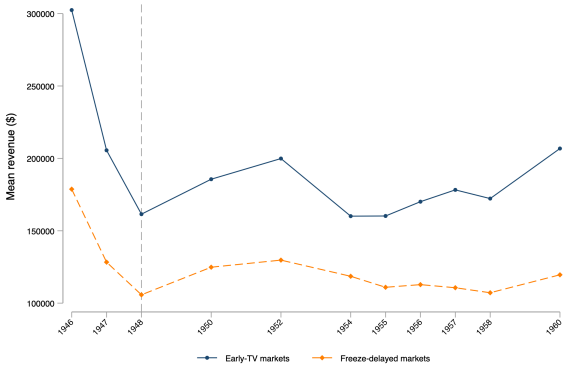
Notes: Panel A plots the cumulative number of U.S. television stations on air by the year in which each station began operation. Panel B reports the number of television stations beginning operation in each year. The dashed vertical line marks the end of World War II. The shaded region denotes the FCC television freeze, which began on September 30, 1948 and was lifted in April 1952. During the freeze, stations that had already received construction permits could proceed toward operation, while the FCC suspended the processing of new and pending applications. The FCC resumed processing applications in July 1952. Each station is counted beginning in the year it commenced television broadcasting.

Figure 4: Timing of First Television Exposure

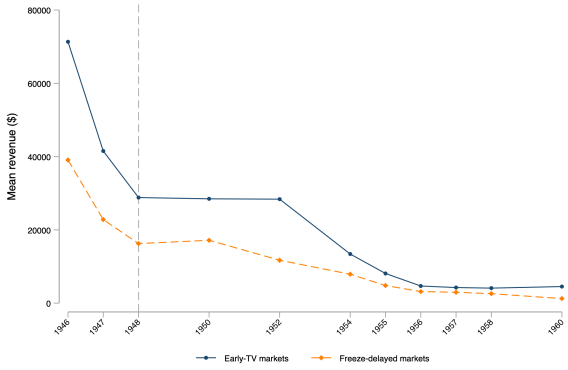


Notes: The figure plots the cumulative share of radio markets receiving their first Grade B television signal, conditional on an identified first television entry that can be classified using the freeze-era application records. Markets already receiving television before September 30, 1948 are excluded. "TV-exposed" markets are those whose first television signal was generated by a station that had received a construction permit before the freeze. "Freeze-delayed" markets are those whose first signal was generated by a station whose application remained pending when the freeze began. If construction-permit and pending stations first reach a market simultaneously, the market is assigned to the TV-exposed group. The figure is constructed for the 866 radio markets for which first television entry can be linked to freeze-era application status, prior to imposing the incumbent-radio and pre-freeze observation restrictions. The vertical dashed lines indicate the beginning and end of the FCC television licensing freeze.

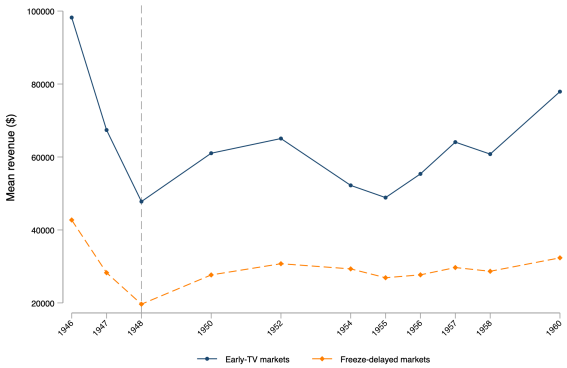
Figure 5: Raw Revenue Trends in Early- and Freeze-Delayed Television Markets



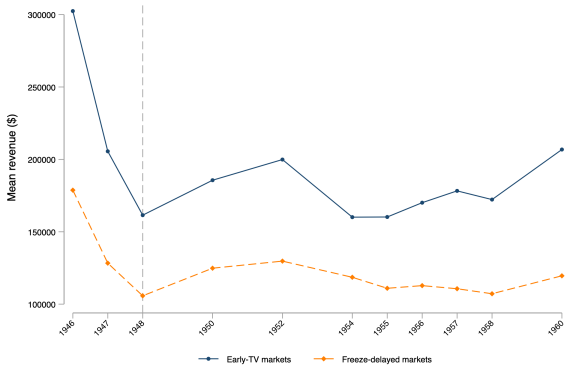
(a) Time-Sales Revenue



(b) Major-Network Revenue



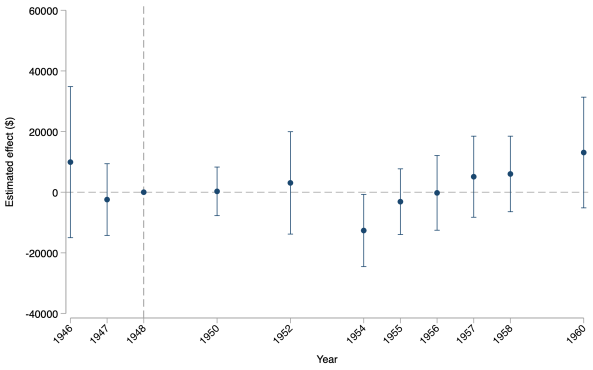
(c) National/Regional Advertising Revenue



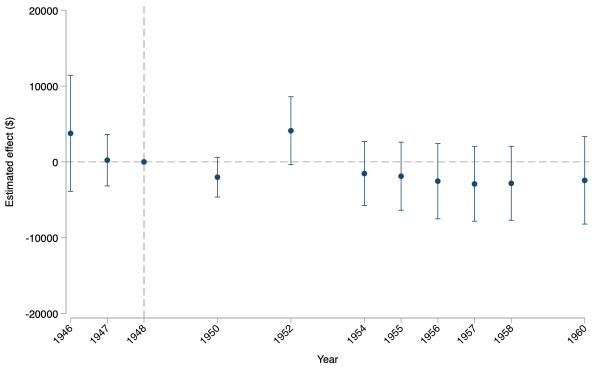
(d) Local Advertising Revenue

Notes: The figure plots unadjusted annual mean revenue for radio stations in the common revenue-analysis sample. The solid line reports means for early-TV markets and the dashed line reports means for freeze-delayed markets. The vertical dashed line marks 1948, when the FCC television licensing freeze began. Panel A reports time-sales revenue, defined as revenue from major, regional, and other network payments together with national/regional and local advertising revenue; Panel B reports major-network revenue; Panel C reports national/regional advertising revenue; and Panel D reports local advertising revenue. Revenue is measured in constant 1950 U.S. dollars. Early-TV and freeze-delayed markets differ in predetermined market characteristics and baseline revenue levels, so the series are descriptive and should not be interpreted as treatment effects. The preferred specifications include radio-station and year fixed effects and allow predetermined 1940 market characteristics to have year-specific effects.

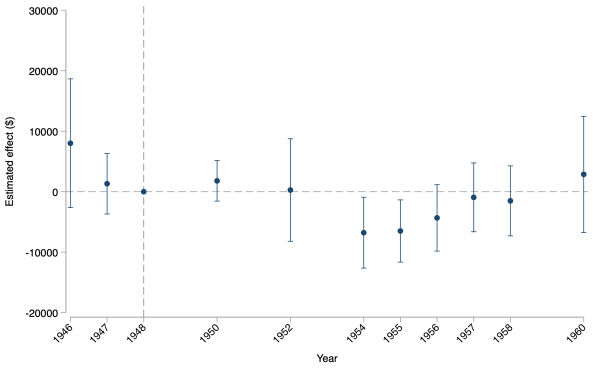
Figure 6: Dynamic Effects of Early Television Exposure on Radio Revenue



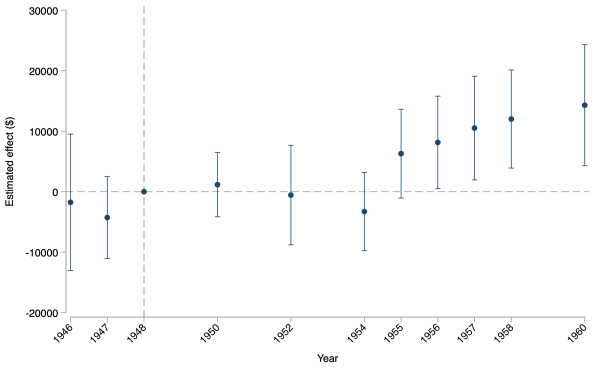
(a) Time-Sales Revenue



(b) Major-Network Revenue



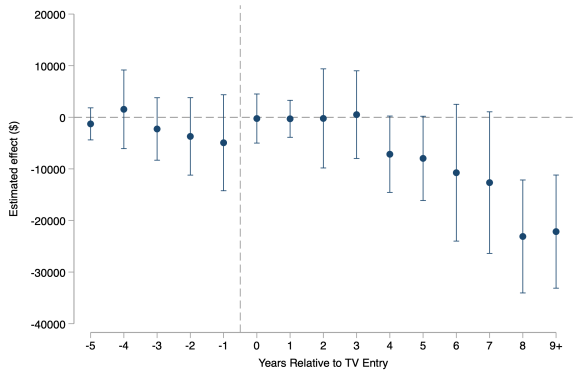
(c) National/Regional Advertising



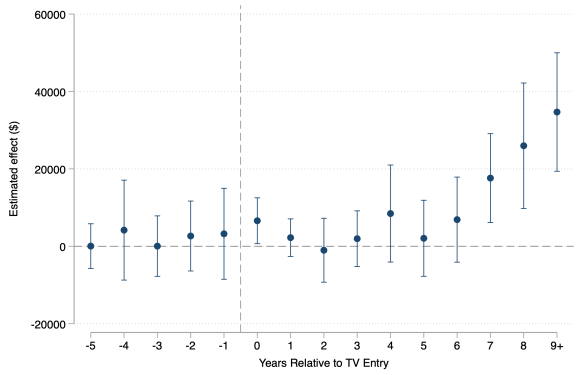
(d) Local Advertising

Notes: The figure reports event-study estimates comparing incumbent radio stations in early-TV markets with stations in freeze-delayed markets, with 1948 omitted as the reference year. All specifications include radio-station and year fixed effects and allow predetermined 1940 market characteristics—log population, urban share, median income, and median years of schooling—to vary by calendar year. Vertical bars report 95 percent confidence intervals based on standard errors clustered by radio market. The vertical dashed line marks 1948, when the FCC television licensing freeze began. Because freeze-delayed markets increasingly receive television after the freeze ends, later coefficients compare markets with earlier versus later television exposure rather than television exposure versus no exposure.

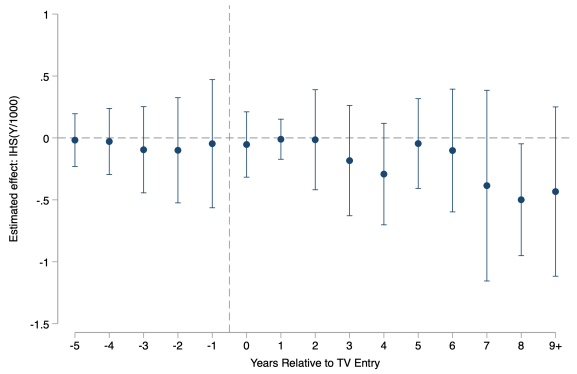
Figure 7: Dynamic Effects of Television Entry on Radio Revenue



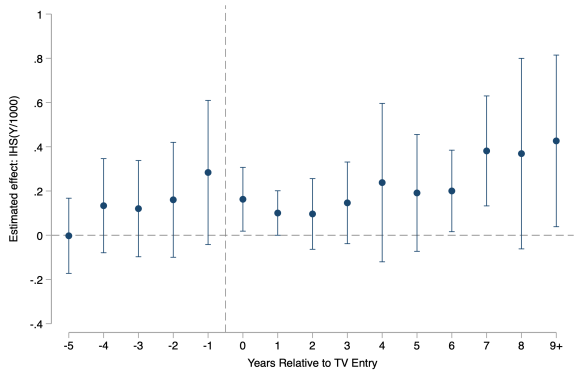
(a) Major-Network Revenue



(b) Local Advertising



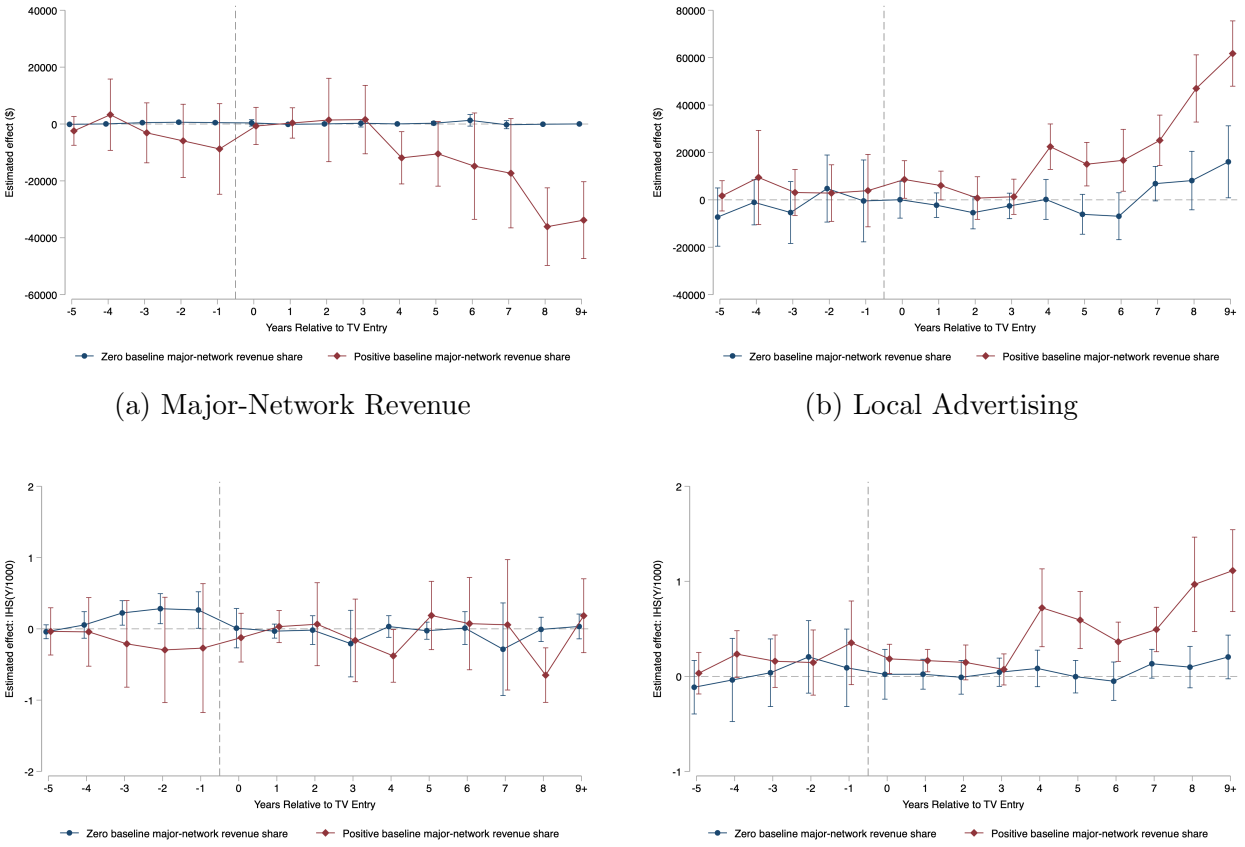
(c) Major-Network Revenue: IHS



(d) Local Advertising: IHS

Notes: The figure reports dynamic treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator for the freeze-based research-design sample. Treatment begins in the calendar year in which a radio market first receives television exposure. The untreated potential-outcome model includes radio-station and year fixed effects. The five pre-treatment coefficients are placebo estimates for years preceding television entry. The top row reports revenue in constant 1950 U.S. dollars, while the bottom row reports inverse hyperbolic sine transformations after expressing revenue in thousands of constant 1950 U.S. dollars. Event years 9–11 are pooled in the final 9+ bin. Vertical bars report 95 percent confidence intervals based on standard errors clustered by radio market using the leave-one-cluster-out adjustment. The vertical dashed line separates the pre-treatment and post-treatment periods.

Figure 8: Dynamic Effects of Television Entry by Baseline Major-Network Revenue Share: BJS Estimates



(a) Major-Network Revenue

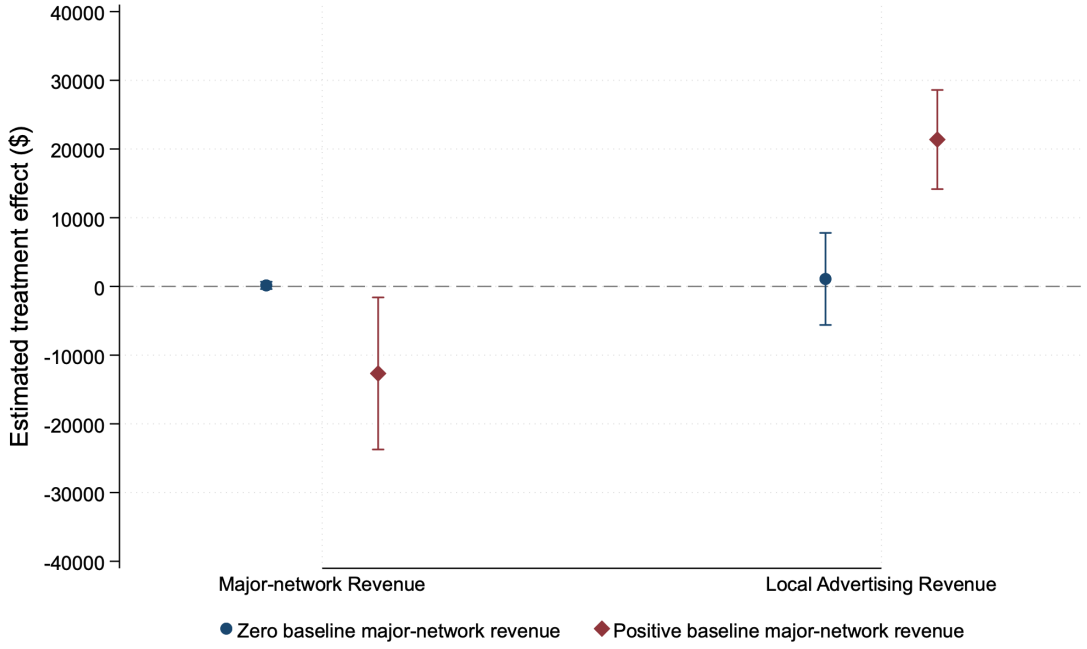
(b) Local Advertising

(c) Major-Network Revenue: IHS

(d) Local Advertising: IHS

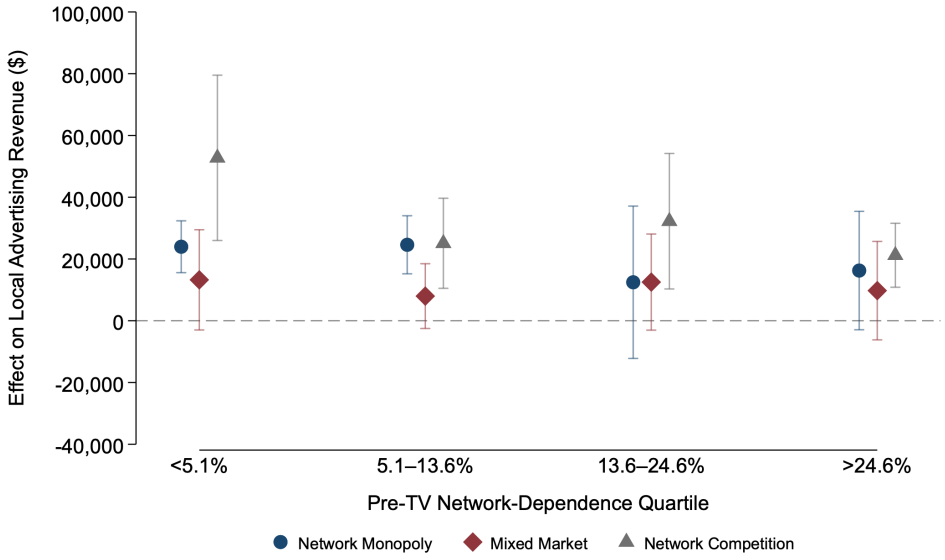
Notes: The figure reports dynamic treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator separately for stations with zero and positive baseline major-network revenue share in the freeze-based research-design sample. Baseline major-network revenue share is measured over 1946–1948 as the ratio of major-network revenue to time-sales revenue across usable baseline observations. Stations with a baseline share equal to zero comprise the zero-share group, while stations with a strictly positive baseline share comprise the positive-share group. Treatment begins in the calendar year in which a radio market first receives television exposure. The untreated potential-outcome model includes radio-station and year fixed effects. The five pre-treatment coefficients are BJS placebo estimates and are estimated directly rather than normalized relative to an omitted event period. The top row reports revenue in constant 1950 U.S. dollars, while the bottom row reports inverse hyperbolic sine transformations after expressing revenue in thousands of constant 1950 U.S. dollars. Event years 9–11 are pooled in the final 9+ bin. Vertical bars report 95 percent confidence intervals based on standard errors clustered by radio market using the leave-one-cluster-out adjustment. The vertical dashed line separates the pre-treatment and post-treatment periods.

Figure 9: Effects of Television Entry by Baseline Major-Network Revenue Dependence



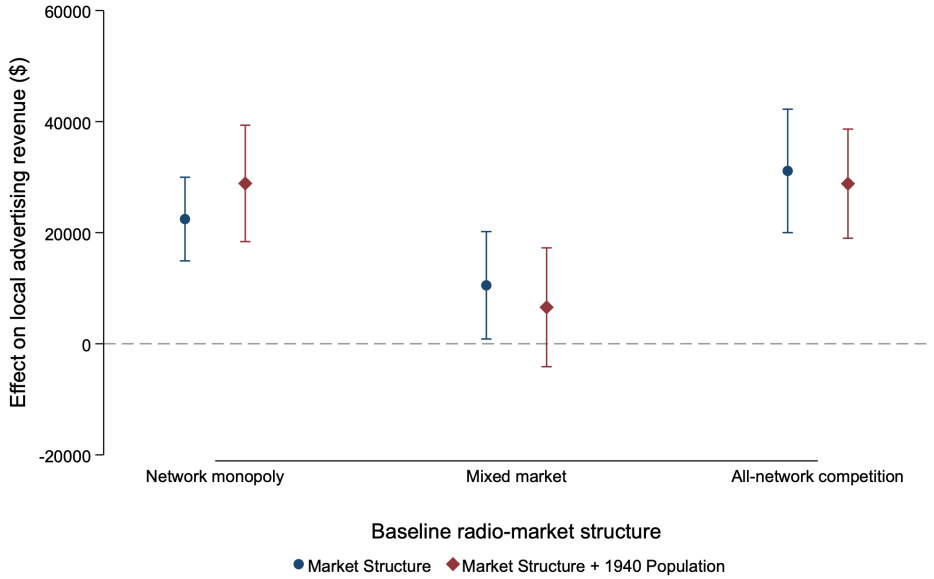
Notes: The figure reports Borusyak–Jaravel–Spiess (BJS) estimates of the effects of television entry on major-network revenue and local advertising revenue, separately for stations with zero and positive baseline major-network revenue. Baseline dependence is measured over 1946–1948. Revenue is expressed in constant 1950 U.S. dollars. Vertical bars represent 95 percent confidence intervals. Standard errors are clustered by radio market.

Figure 10: Heterogeneity in Local Advertising Effects by Baseline Network Dependence and Radio-Market Structure



Notes: The figure reports Borusyak–Jaravel–Spiess (BJS) estimates of the effects of television entry on local advertising revenue, separately by quartile of baseline network dependence and pre-television radio-market structure. Network dependence is measured as the ratio of major-network revenue to time-sales revenue over 1946–1948. Network-monopoly markets contain one incumbent station with a positive observed baseline major-network revenue share and no incumbent with an observed zero share. Mixed markets contain both positive- and zero-share incumbent stations. All-network competition markets contain at least two positive-share incumbents and no observed zero-share incumbent. Network-connected stations are identified by positive baseline major-network revenue. Vertical bars represent 95 percent confidence intervals. Revenue is expressed in constant 1950 U.S. dollars. Standard errors are clustered by radio market.

Figure 11: Local Advertising Effects by Pre-TV Radio-Market Structure



Notes: The figure reports Borusyak–Jaravel–Spiess (BJS) estimates of television-entry effects on local advertising revenue by pre-TV radio-market structure. Market types are defined using 1946–1948 station financial data. Network-monopoly markets have one positive-network-revenue incumbent and no observed zero-revenue incumbent; mixed markets have both positive- and zero-revenue incumbents; all-network competition markets have at least two positive-revenue incumbents and no observed zero-revenue incumbent. Estimates are shown with and without centered log 1940 county population in the treatment-effect projection. Population-adjusted effects are evaluated at the centering value of log population; population does not enter the untreated-outcome model. Revenue is measured in constant 1950 U.S. dollars. Vertical bars represent 95% confidence intervals based on standard errors clustered by radio market. The untreated-outcome model includes station and year fixed effects.

A Conceptual Framework: Comparative Statics and Derivations

This appendix gives the derivations for the conceptual framework in Section 3. A national network supplies a radio station with a bundle of sponsored programs and advertisements. The station takes that bundle as given and decides how to use the rest of its broadcast schedule. I first describe why the network may reduce its bundle when television enters the local market, then examine the station's advertising choices.

A.1 Network Programming and Television

The network sells advertising alongside its programs and pays the station to broadcast them. It can supply a program to radio and television at the same time. The decision that matters here is whether it continues supplying a sponsored program to this particular radio station.

Consider a sponsored program j that uses $h_j > 0$ of the station's broadcast schedule. Let $T \in [0, 1]$ measure television exposure in the local market. Write $V_j(T)$ for the extra profit the network earns from supplying this program to the station, after accounting for advertising revenue and the payment to the station. The network's other distribution decisions, including television broadcasts, are held fixed. It continues supplying program j when

$$V_j(T) \geq 0. \tag{9}$$

I assume that $V_j(T)$ weakly decreases as television becomes available locally. Television gives advertisers another way to reach people in the market and may reduce what the network earns from supplying the same sponsored program through radio. The network can stop supplying this station without withdrawing the program from other markets.

Let $n(T)$ be the share of the station's schedule occupied by the network bundle, including the network's advertisements. Adding the programs the network continues to supply gives

$$n(T) = \sum_j h_j \mathbf{1}\{V_j(T) \geq 0\}. \tag{10}$$

Under this assumption, $n(T)$ cannot increase when local television exposure rises. I study the case in which at least one sponsored program is withdrawn, so

$$n_1 < n_0, \tag{11}$$

where subscripts 0 and 1 refer to the periods before and after the increase in television

exposure.

If the station receives $w_j > 0$ per unit of airtime for program j, its network revenue is

$$R_{\text{net}}(T) = \sum_j w_j h_j \mathbf{1}\{V_j(T) \geq 0\}. \tag{12}$$

When the network withdraws a sponsored program, the station loses its payment but gains control of the airtime. Network revenue falls if payments for the programs that remain do not change. I do not model changes in those payments.

This is a decision affecting an individual affiliate. Networks could also change programming nationwide, but changes common to all stations in the same year are absorbed by year fixed effects in the empirical analysis. The model focuses on the additional change associated with television entering the station's own market.

A.2 The Radio Station's Schedule and Advertising Demand

I normalize the station's broadcast day to one. The network bundle occupies n, the station's own programs occupy s, and advertising sold directly by the station occupies a. Advertisements sold by the network are included in n, not a. Thus,

$$n + s + a = 1. \tag{13}$$

A station needs programming to support the advertisements it sells. I assume that it can sell at most $\chi > 0$ units of advertising for each unit of its own programming:

$$a \leq \chi s, \qquad \chi > 0. \tag{14}$$

Using $s = 1 - n - a$, this becomes

$$a \leq \frac{\chi(1-n)}{1+\chi}. \tag{15}$$

The right-hand side is the most advertising time the station can sell for a given network bundle. Losing network airtime increases this capacity, but the station will use it only if advertiser demand is sufficient.

The station sells advertising to local businesses and to national and regional advertisers. Let q_L and q_N be the amounts of advertising time purchased by these groups. Together they account for the station's advertising sales:

$$a = q_L + q_N. \tag{16}$$

Both groups buy time using the station's spot rate card. I represent the rate they face by a common effective price, p. Their demands are

$$\begin{aligned} q_L &= b\left[(v_L - \tau_L T)(1-a) + \rho s - p\right], \\ q_N &= b\left[(v_N - \tau_N T)(1-a) - p\right], \end{aligned} \tag{17}$$

where $b > 0$ measures the response of demand to price. The term $1-a = s+n$ is the part of the schedule not sold directly by the station. It includes the station's programs and the network bundle, including network advertisements. It captures how the available broadcast schedule enters advertising demand; it does not measure the station's number of listeners or predict how listenership changes.

The parameters v_L and v_N measure how much each advertiser group values radio before television. Television reduces these values by $\tau_L T$ and $\tau_N T$. I assume $\tau_N > \tau_L \geq 0$: national and regional advertisers have more reason to switch to television than local businesses. The values $v_L - \tau_L T$ and $v_N - \tau_N T$ are assumed positive over the range considered.

The term ρs, where $\rho > 0$, gives the station's own programming additional value to local advertisers. For example, local programming may help a business reach customers interested in its products. The model does not determine which programs the station chooses or how many people listen.

Adding the two demand equations and using $a = q_L + q_N$ gives

$$a = b\left[(v_L + v_N - (\tau_L + \tau_N)T)\,(1-a) + \rho s - 2p\right]. \tag{18}$$

Solving for the spot rate yields

$$p = \frac{1}{2}\left[(v_L + v_N - (\tau_L + \tau_N)T)\,(1-a) + \rho s - \frac{a}{b}\right]. \tag{19}$$

Holding the schedule and advertiser valuations fixed, a larger quantity of station-sold advertising comes with a lower rate. More station-controlled programming can increase what local advertisers are willing to pay. I assume both groups continue buying positive quantities and that $p > 0$. Otherwise, the demand of a group that stops buying would be set to zero, and the price equation would change.

A.3 The Station's Optimal Advertising Choice

The network chooses n. The station then chooses advertising time a, which leaves $s = 1-n-a$ for its own programs. Since network payments do not depend on this choice, the station

solves

$$\max_{0\leq a\leq \frac{\chi(1-n)}{1+\chi}} \left\{p(a,n,T)a + R_{\text{net}}(T)\right\}. \tag{20}$$

Substituting $s = 1 - n - a$ into the price equation gives

$$p(a,n,T) = \frac{1}{2}\left[\left(v_L + v_N - (\tau_L + \tau_N)T\right)(1-a) + \rho(1-n-a) - \frac{a}{b}\right]. \tag{21}$$

The first-order condition without the advertising-capacity constraint is

$$\begin{aligned} 0 = \frac{\partial(pa)}{\partial a} = \frac{1}{2}\Bigg[& \left(v_L + v_N - (\tau_L + \tau_N)T\right) + \rho(1-n) \\ & - 2\left(v_L + v_N - (\tau_L + \tau_N)T + \rho + \frac{1}{b}\right)a\Bigg]. \end{aligned} \tag{22}$$

The second derivative is

$$\frac{\partial^2(pa)}{\partial a^2} = -\left(v_L + v_N - (\tau_L + \tau_N)T + \rho + \frac{1}{b}\right) < 0. \tag{23}$$

Thus the station has a unique unconstrained advertising choice:

$$a^{\text{unc}} = \frac{\left(v_L + v_N - (\tau_L + \tau_N)T\right) + \rho(1-n)}{2\left(v_L + v_N - (\tau_L + \tau_N)T + \rho + \frac{1}{b}\right)}. \tag{24}$$

With the advertising limit, its choice is

$$a^* = \min\left\{a^{\text{unc}}, \frac{\chi(1-n)}{1+\chi}\right\}. \tag{25}$$

I focus on stations with enough advertiser demand to sell all the commercial time their programming can support. The condition is

$$a^{\text{unc}} \geq \frac{\chi(1-n)}{1+\chi} \tag{26}$$

before and after television entry. Under this assumption the advertising limit binds, and the station divides the time left by the network between its own programming and advertising:

$$a^* = \frac{\chi(1-n)}{1+\chi}, \qquad s^* = \frac{1-n}{1+\chi}. \tag{27}$$

Writing $\Delta x = x_1 - x_0$ for the change after television entry, a reduction in the network

bundle gives

$$\Delta s = -\frac{\Delta n}{1+\chi} > 0, \qquad \Delta a = -\frac{\chi \Delta n}{1+\chi} > 0. \tag{28}$$

For every unit of airtime released by the network, the station uses $1/(1+\chi)$ for its own programs and $\chi/(1+\chi)$ for commercials. If the advertising limit does not bind, more available time need not lead to more advertising sales.

A.4 Advertising Rates

I now examine how the spot rate changes when television exposure rises from $T = 0$ to $T = 1$ and the network reduces its bundle. From Equation 19,

$$2\Delta p = -(v_L + v_N)\Delta a - (\tau_L + \tau_N)(1 - a_1) + \rho \Delta s - \frac{\Delta a}{b}. \tag{29}$$

Selling more commercials lowers the rate through the first and last terms. Television also reduces advertiser demand, captured by the second term. More station-controlled programming can offset some of these effects because local advertisers value it, as shown by the third term. None of these terms is a measured change in listenership.

Under the binding advertising limit, $\Delta a = \chi \Delta s$. Substituting gives

$$2\Delta p = -(\tau_L + \tau_N)(1 - a_1) + \left[\rho - \chi\left(v_L + v_N + \frac{1}{b}\right)\right] \Delta s. \tag{30}$$

A sufficient condition for the rate to fall is

$$\rho \leq \chi\left(v_L + v_N + \frac{1}{b}\right). \tag{31}$$

This says that the additional value of the station's own programming to local advertisers is not large enough to offset the downward pressure from selling more advertisements. The rate may still fall without this condition if television reduces demand sufficiently.

A.5 Local and National/Regional Advertising

Local and national/regional advertisers buy from the same station at the same spot rate, but they may want different amounts of advertising after television arrives. Subtracting their demand equations removes the common price:

$$\frac{q_L - q_N}{b} = \left[v_L - v_N + (\tau_N - \tau_L)T\right](1 - a) + \rho s. \tag{32}$$

For a change from $T = 0$ to $T = 1$,

$$\frac{\Delta(q_L - q_N)}{b} = -(v_L - v_N)\Delta a + (\tau_N - \tau_L)(1 - a_1) + \rho\Delta s. \tag{33}$$

The second term is positive because national and regional advertisers have more reason to switch to television. The third is positive because station-controlled programming expands and is particularly useful to local advertisers. The first term depends on which group valued radio more before television.

One sufficient condition for the difference in quantities to shift toward local advertising is

$$v_N \geq v_L. \tag{34}$$

With $\Delta a > 0$, this condition makes the first term nonnegative. The other two terms are positive, so

$$\Delta(q_L - q_N) > 0. \tag{35}$$

Since

$$q_L = \frac{a + (q_L - q_N)}{2}, \qquad q_N = \frac{a - (q_L - q_N)}{2}, \tag{36}$$

and total station-sold advertising increases, local advertising quantity also rises:

$$\Delta q_L > 0. \tag{37}$$

The national and regional response is less clear:

$$2\Delta q_N = \Delta a - \Delta(q_L - q_N). \tag{38}$$

It may increase if the station sells enough additional airtime or decrease if the shift toward local demand is larger. The result for local advertising quantity depends on the assumptions above; it is not implied by network withdrawal alone.

A.6 Advertising Revenue and the Limits to Replacement

Selling more local advertising does not automatically increase local advertising revenue. The revenue from local and national/regional spot advertising is

$$R_L = pq_L, \tag{39}$$

and $R_N = pq_N$. The change in local revenue is

$$\Delta R_L = p_1 \Delta q_L + q_{L0} \Delta p. \tag{40}$$

The first term is the revenue from additional advertising sold at the new rate. The second is the change in revenue earned on the original amount of advertising. If the rate falls, local revenue rises only when

$$p_1 \Delta q_L > -q_{L0} \Delta p. \tag{41}$$

For positive initial quantities and prices, this is equivalent to

$$\frac{q_{L1}}{q_{L0}} > \frac{p_0}{p_1}. \tag{42}$$

The station must sell enough additional local advertising to make up for the lower rate. Similarly, total station-sold spot revenue is $R_{\text{spot}} = pa$, with change

$$\Delta R_{\text{spot}} = p_1 \Delta a + a_0 \Delta p. \tag{43}$$

This can be positive or negative. The same is true of national and regional spot revenue.

The station also has limited time to sell. Since $n \geq 0$, Equation 27 implies

$$a^* \leq \frac{\chi}{1+\chi}. \tag{44}$$

Local advertising quantity cannot exceed this limit. With positive prices, nonnegative television exposure, and the assumed positive advertiser values, the spot-rate equation also gives the upper bound

$$p \leq \frac{v_L + v_N + \rho}{2}. \tag{45}$$

Combining the two bounds gives

$$R_L \leq \frac{\chi}{2(1+\chi)}(v_L + v_N + \rho). \tag{46}$$

This is a loose bound, not a prediction that local revenue approaches it. It simply shows that releasing network airtime cannot generate unlimited local advertising revenue. There is only so much time to sell, and advertisers' willingness to pay is finite. The model does not imply that each additional reduction in network airtime produces a smaller local revenue gain.

A.7 Station-Level Revenue and the Zero-Network Case

Total station time-sales revenue in the model is the sum of major-network payments and advertising sold directly by the station:

$$R^{TS} = R_{\text{net}} + R_L + R_N. \tag{47}$$

Thus,

$$\Delta R^{TS} = \Delta R_{\text{net}} + \Delta R_L + \Delta R_N. \tag{48}$$

Network payments may fall while local advertising revenue rises. National and regional revenue can move either way. The total effect on station time-sales revenue is therefore ambiguous. The empirical time-sales measure also includes regional and other network payments, which are not modeled separately.

Consider the special case of a station that receives no network bundle before or after television entry, so $n_0 = n_1 = 0$. If the advertising limit binds in both periods, the station's programming and spot advertising time are fixed:

$$s^* = \frac{1}{1+\chi}, \qquad a^* = \frac{\chi}{1+\chi}. \tag{49}$$

Television does not free additional network airtime in this case. It can still reduce advertisers' willingness to pay for radio. Holding s and a fixed, the common spot rate changes by

$$\Delta p = -\frac{(\tau_L + \tau_N)(1-a)}{2} < 0. \tag{50}$$

If both advertiser groups continue buying advertising, their quantities change by

$$\Delta q_L = \frac{b(\tau_N - \tau_L)(1-a)}{2} > 0, \qquad \Delta q_N = -\Delta q_L. \tag{51}$$

Local advertising quantity increases at the expense of national and regional advertising in this special case, while total advertising time stays fixed. Whether local revenue increases depends on the falling spot rate.

This special case is narrower than the zero-network-revenue group in the data. A station with no reported payments from ABC, CBS, NBC, or MBS may still receive network programs or earn revenue through other arrangements. The model does not explain every response of those stations, including their estimated increase in national and regional spot revenue.

A.8 Scope of the Model

The main results require a reduction in the network bundle, a binding advertising limit, and positive demand from both advertiser groups. Under these assumptions, the station increases its own programming and sells more advertising time. The amount of time not sold directly by the station, $1 - a$, becomes smaller, but this says nothing definite about the number or types of listeners. The model does not determine how audiences respond when programs change.

The spot rate falls under Equation 31. Local advertising quantity increases under the additional condition in Equation 34, while national and regional advertising quantity may rise or fall. Local revenue rises only if the quantity increase makes up for any reduction in the rate. The model therefore does not require every radio station to gain local revenue or replace lost network payments. Nor does it predict that stations with larger initial network businesses must gain more local advertising revenue.

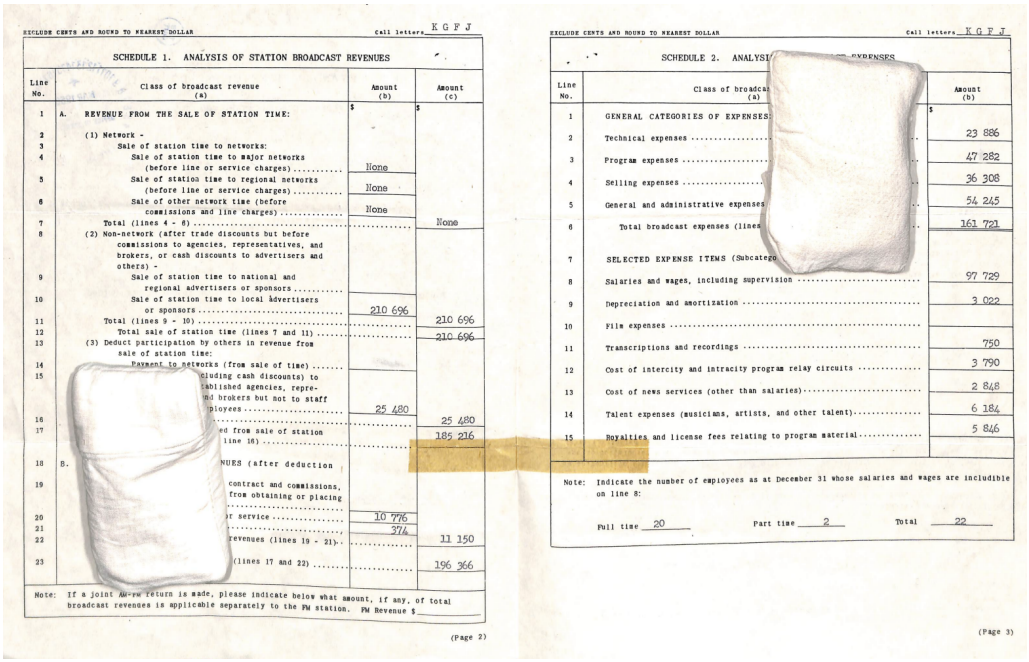
The station chooses programming time, not the type of program or its cost. It could spend less on programming even if it broadcasts more of its own programs. The model also has only one radio station, so it does not explain competition for advertisers between stations. Those questions remain outside the $J = 1$ framework.

B Data Appendix

B.1 Example Archival Record

Appendix Figure 12 provides an example of the archival FCC station financial records used to construct the American Radio Panel.

Figure 12: Example of the Archival Records Underlying the American Radio Panel



EXCLUDE CENTS AND ROUND TO NEAREST DOLLAR — Call letters K G F J

SCHEDULE 1. ANALYSIS OF STATION BROADCAST REVENUES

Line No.	Class of broadcast revenue (a)	Amount (b)	Amount (c)
1	A. REVENUE FROM THE SALE OF STATION TIME:	$	$
2	(1) Network -		
3	Sale of station time to networks:		
4	Sale of station time to major networks (before line or service charges)	None	
5	Sale of station time to regional networks (before line or service charges)	None	
6	Sale of other network time (before commissions and line charges)	None	
7	Total (lines 4 - 6)		None
8	(2) Non-network (after trade discounts but before commissions to agencies, representatives, and brokers, or cash discounts to advertisers and others) -		
9	Sale of station time to national and regional advertisers or sponsors		
10	Sale of station time to local advertisers or sponsors	210 696	
11	Total (lines 9 - 10)		210 696
12	Total sale of station time (lines 7 and 11)		210 696
13	(3) Deduct participation by others in revenue from sale of station time:		
14	Payment to networks (from sale of time)		
15	...luding cash discounts) to ...tablished agencies, repre- ...nd brokers but not to staff ...ployees	25 480	
16			25 480
17	...ed from sale of station line 16)		185 216
18	B. ...NUES (after deduction		
19	... contract and commissions, from obtaining or placing		
20	...r service	10 776	
21		374	
22	...revenues (lines 19 - 21)		11 150
23	...(lines 17 and 22)		196 366

Note: If a joint AM-FM return is made, please indicate below what amount, if any, of total broadcast revenues is applicable separately to the FM station. FM Revenue $

(Page 2)

EXCLUDE CENTS AND ROUND TO NEAREST DOLLAR — Call letters K G F J

SCHEDULE 2. ANALYSI... EXPENSES

Line No.	Class of broadca... (a)	Amount (b)
1	GENERAL CATEGORIES OF EXPENSES:	$
2	Technical expenses	23 886
3	Program expenses	47 282
4	Selling expenses	36 308
5	General and administrative expenses	54 245
6	Total broadcast expenses (lines ...	161 721
7	SELECTED EXPENSE ITEMS (Subcatego...	
8	Salaries and wages, including supervision	97 729
9	Depreciation and amortization	3 022
10	Film expenses	
11	Transcriptions and recordings	750
12	Cost of intercity and intracity program relay circuits	3 790
13	Cost of news services (other than salaries)	2 848
14	Talent expenses (musicians, artists, and other talent)	6 184
15	Royalties and license fees relating to program material	5 846

Note: Indicate the number of employees as at December 31 whose salaries and wages are includible on line 8:

Full time 20 Part time 2 Total 22

(Page 3)

Notes: The figure shows an example of the archival FCC station financial records used to construct the American Radio Panel. The original records were professionally hand-transcribed and subsequently cleaned, standardized, and linked across years and to contemporaneous editions of the *Broadcasting Yearbook* to construct the longitudinal station-year panel used in the analysis. *Source:* Federal Communications Commission station financial records.

B.2 Construction of Local Television Exposure

I construct local television exposure using two sources of information: the regulatory status of proposed television stations at the beginning of the FCC licensing freeze and the subsequent timing and geographic coverage of television broadcasts. The freeze-era records identify whether a television station had received a construction permit before September 30, 1948 or had an application still pending when the FCC suspended processing. Operating dates and signal coverage determine when television first became available in each radio market. I use the regulatory information to classify markets into early and freeze-delayed groups and the broadcast information to assign realized television-entry dates.

I construct the freeze-era television-station database from the January 1949 *Television Rates & Factbook: TV Directory No. 6*, published by the Radio News Bureau (Radio News Bureau, 1949). I digitized the relevant listings and manually verified them against the original directory. The records report proposed television stations' cities and states, call letters when available, and construction-permit or pending-application status at the beginning of the freeze. Stations that had received construction permits could proceed toward operation during the suspension, while applications still pending could not receive authorization until licensing resumed. Koenig (2023) also uses this directory to identify locations affected by the FCC freeze.

I link the application records to the television-station data constructed by Angelucci et al. (2024), which report stations' first broadcast dates, transmitter locations, and Grade B signal contours. I use the first broadcast date to measure television entry and the signal contours to identify locations receiving television. I consider all television stations in the coverage data, including those absent from the January 1949 directory, when determining the first signal reaching a radio market. This accounts for cases in which another television station began serving a market before the station identified in the freeze-era records. I retain a market in the freeze-based research-design sample only when the station responsible for its first television exposure can be assigned a freeze-era regulatory status.

I assign television exposure using the geographic location of each radio market. I geocode radio stations' cities of license using city, county, and state information from the American Radio Panel and use the city centroid as the receiver location. I then compare each centroid with the Grade B coverage contours of operating television stations. A market is classified as receiving television when its centroid falls within the coverage area of an operating station. Among all television stations whose signals reach the market, I use the earliest broadcast date to determine the year of first exposure. I exclude markets that were already receiving television before September 30, 1948.

I match the freeze-era applications to operating television stations using call letters and

city and state identifiers. Some pending applications do not report call letters. For these observations, I use the proposed city and state to link the application to subsequent television operation and assign the first station beginning operation in that location after the freeze to the pending application. I also use city, county, and state identifiers to reconcile television locations with the radio panel and historical Census data. When stations with different regulatory statuses begin providing the first television signal to a market on the same date, I classify the market as early TV if at least one of those stations had received a construction permit before the freeze.

I construct two treatment measures from these matches. The benchmark early-versus-delayed specification classifies a market as *early TV* when the station responsible for its first television exposure had received a construction permit before September 30, 1948. A market is classified as *freeze delayed* when the station responsible for its first exposure had an application pending when the freeze began and television did not become available until after licensing resumed. This classification uses the regulatory status of the station providing the first local television signal, rather than assigning treatment groups solely from the observed year of television entry.

The staggered-adoption specification instead uses the year in which a radio market first receives a television signal. A construction permit does not itself constitute television exposure, and the end of the freeze does not automatically assign treatment to markets with pending applications. Treatment begins when an operating television station's signal reaches the market centroid. For markets first receiving television late in 1952, I assign treatment to the 1953 cohort so that the treatment year corresponds to the first annual reporting period with meaningful exposure. I retain freeze-delayed markets that have a qualifying pending application but do not receive television during the estimation window as untreated comparison markets.

The two specifications use different sources of variation in television timing. The benchmark comparison uses the regulatory status of the station providing first exposure, which was determined before the subsequent expansion of television. The staggered-adoption specification uses realized entry dates, which more closely measure when radio stations faced television competition but may also reflect subsequent station decisions and local economic conditions. The freeze-based sample is itself defined using realized first exposure, so the benchmark is not an unconditional assignment based solely on application status in September 1948. I therefore interpret the early-versus-delayed estimates as a reduced-form comparison based on the regulatory interruption, while the staggered estimates measure responses relative to realized television entry. The identification assumptions for each specification are discussed in Section 5.

C Sample Construction

Figure 13 summarizes the construction of the common incumbent-radio research-design population. Appendix Table 12 reports the corresponding sample counts and the additional baseline-observed restriction used when pre-treatment station characteristics are required.

Figure 13: Construction of the Main Incumbent Radio Sample

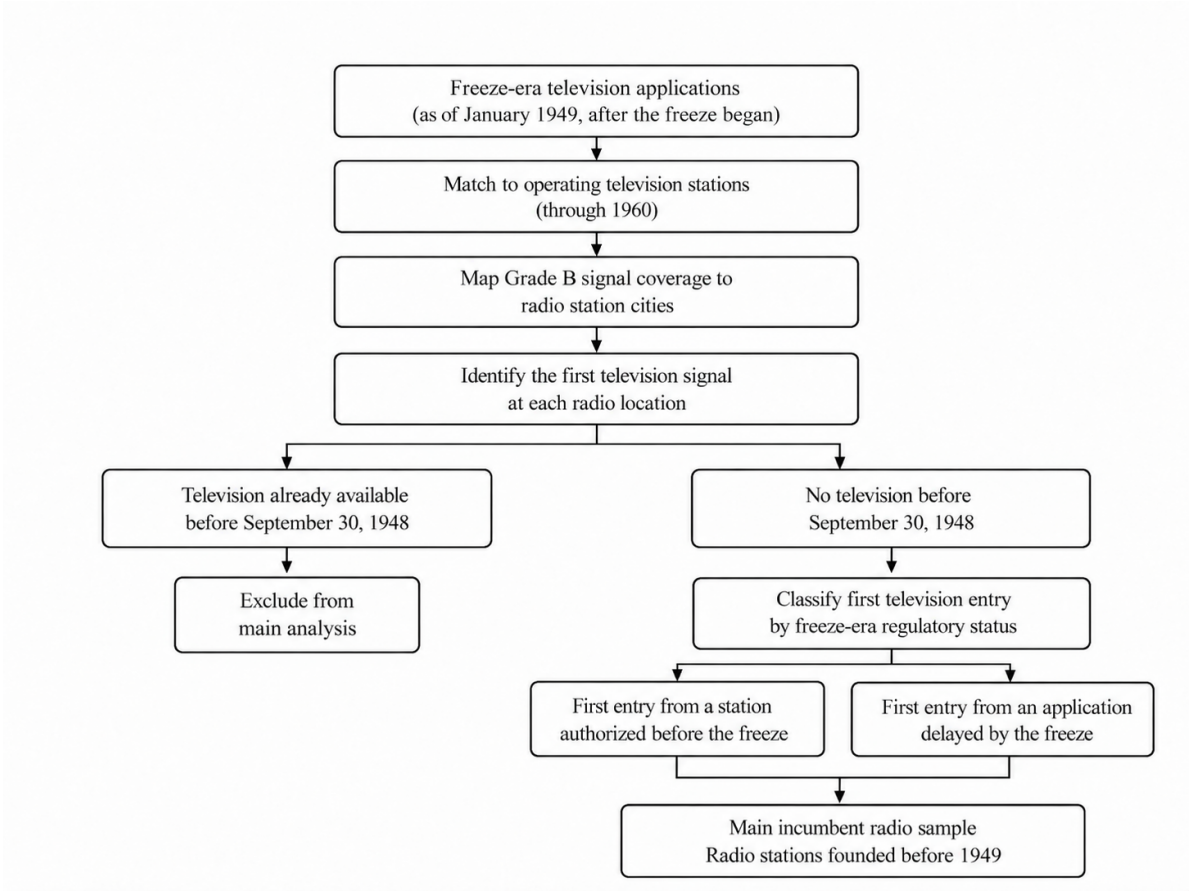
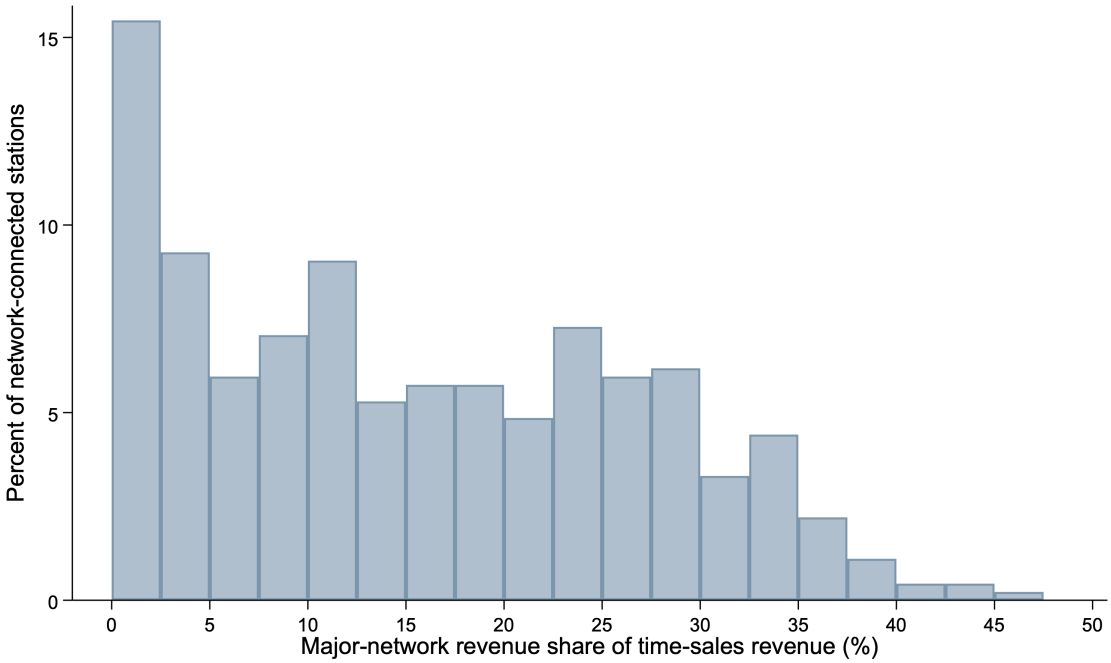


Table 12: Sample Construction

Restriction	Radio Stations	Radio Markets	Station-years
Full American Radio Panel	4,324	2,297	29,456
First television entry identified	3,955	2,035	27,017
Exclude radio markets with television before Sept. 30, 1948	2,755	1,500	18,923
First TV entrant matched to freeze-era application status	1,701	866	11,708
Restrict to pre-freeze incumbent radio stations	902	466	8,176
Require at least one observation in 1946–1948	794	458	7,660

Notes: Restrictions are applied sequentially. Radio markets correspond to the city locations of radio stations in the American Radio Panel. The first television entrant is classified using its status in the FCC freeze-era application records. Pre-freeze incumbent radio stations are stations founded before 1949. Restricting to these stations yields the common incumbent-radio research-design population. The final row additionally requires that a station be observed at least once during 1946–1948, before the divergence in television exposure induced by the FCC freeze, and therefore defines the baseline-observed subsample used in analyses requiring pre-treatment station characteristics. Estimation samples may be smaller because of outcome availability and estimator-specific requirements, and the corresponding observation counts are reported in each results table.

Figure 14: Distribution of Baseline Major-Network Revenue Dependence



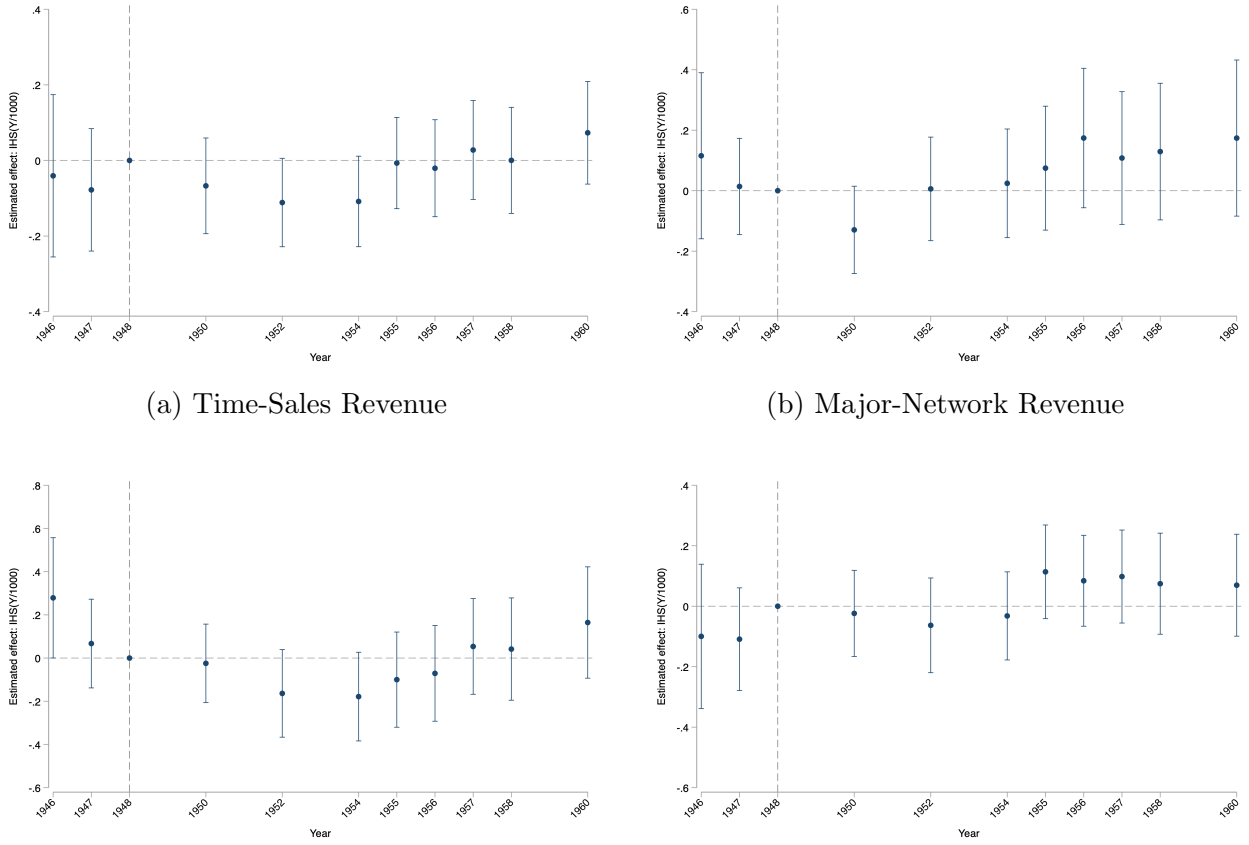
Notes: The figure shows the distribution of baseline major-network revenue dependence among radio stations with positive pre-television major-network revenue. Baseline dependence is measured as major-network revenue divided by time-sales revenue over 1946–1948.

D Additional Empirical Results and Robustness

D.1 Early Television Exposure: IHS Event Studies

Figure 15 reports the IHS counterparts to the level event studies shown in Figure 6.

Figure 15: Dynamic Effects of Early Television Exposure on Radio Revenue: IHS Specifications



(a) Time-Sales Revenue

(b) Major-Network Revenue

(c) National/Regional Advertising

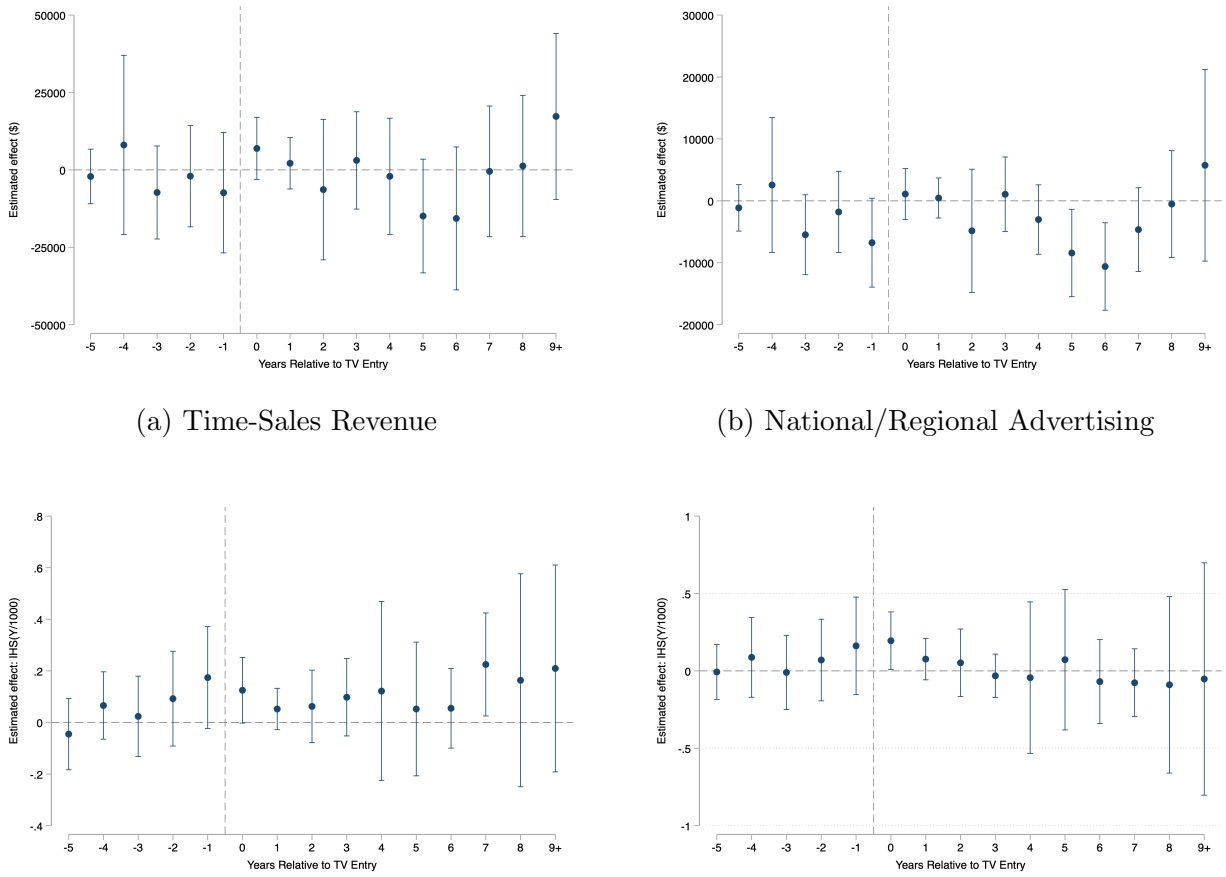
(d) Local Advertising

Notes: The figure reports event-study estimates comparing incumbent radio stations in early-TV markets with stations in freeze-delayed markets, with 1948 omitted as the reference year. Revenue outcomes are transformed using the inverse hyperbolic sine after expressing revenues in thousands of constant 1950 U.S. dollars. All specifications include radio-station and year fixed effects and allow predetermined 1940 market characteristics—log population, urban share, median income, and median years of schooling—to vary by calendar year. Vertical bars report 95 percent confidence intervals based on standard errors clustered by radio market. The vertical dashed line marks 1948, when the FCC television licensing freeze began.

D.2 Additional Dynamic Effects

Appendix Figure 16 reports the corresponding dynamic estimates for time-sales revenue and national/regional advertising revenue. These outcomes complement the major-network and local-advertising responses reported in Figure 7.

Figure 16: Additional Dynamic Effects of Television Entry on Radio Revenue



(a) Time-Sales Revenue

(b) National/Regional Advertising

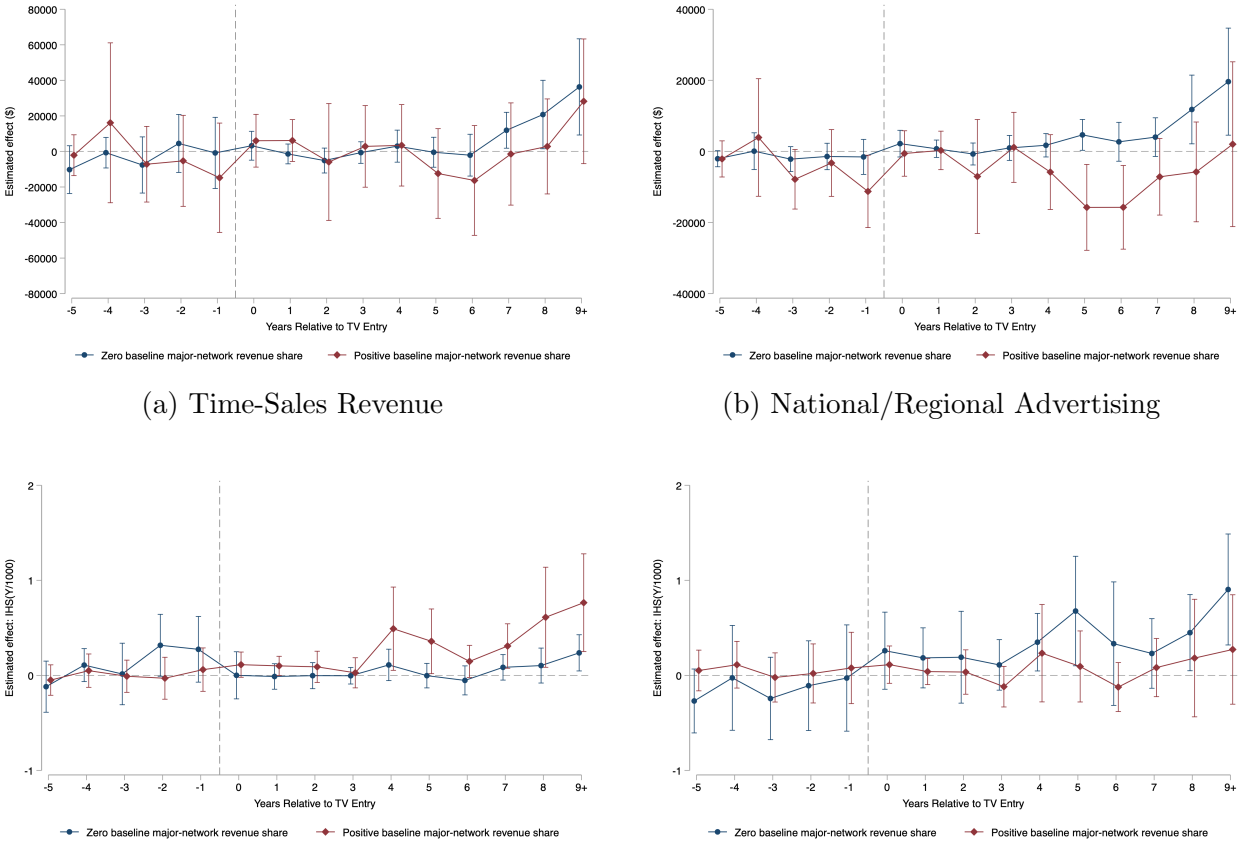
(c) Time-Sales Revenue: IHS

(d) National/Regional Advertising: IHS

Notes: The figure reports dynamic treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator for the freeze-based research-design sample. Treatment begins in the calendar year in which a radio market first receives television exposure. The untreated potential-outcome model includes radio-station and year fixed effects. The five pre-treatment coefficients are placebo estimates for years preceding television entry. The top row reports revenue in constant 1950 U.S. dollars, while the bottom row reports inverse hyperbolic sine transformations after expressing revenue in thousands of constant 1950 U.S. dollars. Time-sales revenue includes payments from major, regional, and other networks as well as national/regional and local advertising revenue. Event years 9–11 are pooled in the final 9+ bin. Vertical bars report 95 percent confidence intervals based on standard errors clustered by radio market using the leave-one-cluster-out adjustment. The vertical dashed line separates the pre-treatment and post-treatment periods.

Figure 17 reports the remaining heterogeneous dynamic estimates for time-sales revenue and national/regional advertising revenue reported in columns (1) and (3) of Table 5.

Figure 17: Additional Dynamic Effects by Baseline Major-Network Revenue Share: BJS Estimates



(a) Time-Sales Revenue

(b) National/Regional Advertising

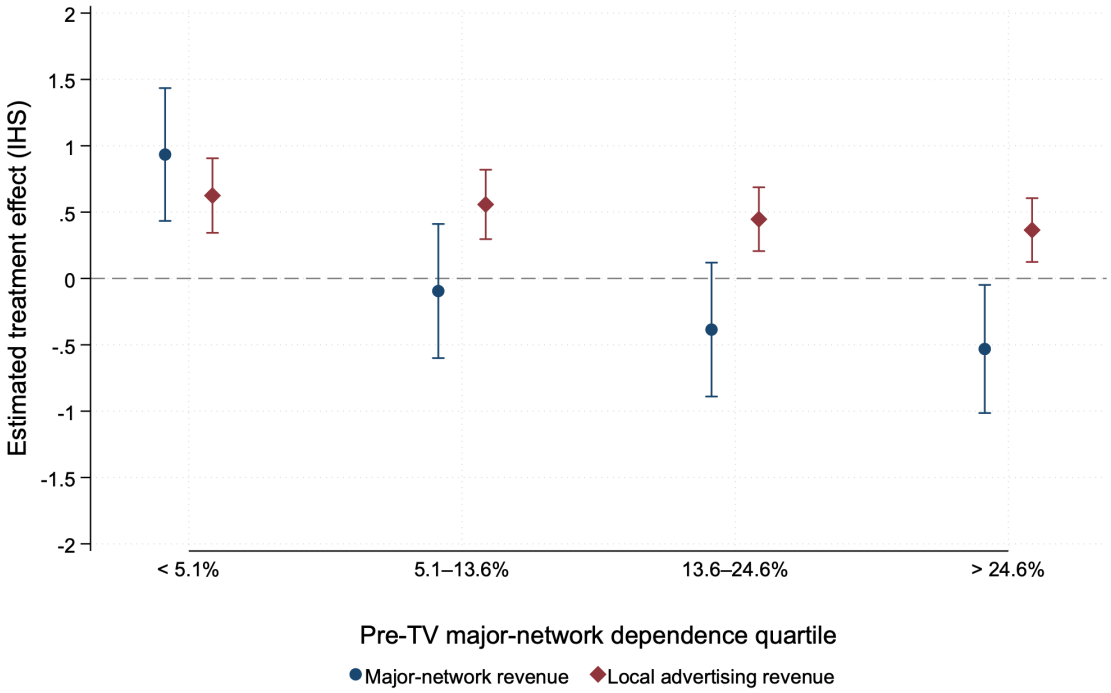
(c) Time-Sales Revenue: IHS

(d) National/Regional Advertising: IHS

Notes: The figure reports dynamic treatment effects estimated using the Borusyak–Jaravel–Spiess imputation estimator separately for stations with zero and positive baseline major-network revenue share in the freeze-based research-design sample. Baseline major-network revenue share is measured over 1946–1948 as the ratio of major-network revenue to time-sales revenue across usable baseline observations. Stations with a baseline share equal to zero comprise the zero-share group, while stations with a strictly positive baseline share comprise the positive-share group. Treatment begins in the calendar year in which a radio market first receives television exposure. The untreated potential-outcome model includes radio-station and year fixed effects. The five pre-treatment coefficients are BJS placebo estimates and are estimated directly rather than normalized relative to an omitted event period. The top row reports revenue in constant 1950 U.S. dollars, while the bottom row reports inverse hyperbolic sine transformations after expressing revenue in thousands of constant 1950 U.S. dollars. Event years 9–11 are pooled in the final 9+ bin. Vertical bars report 95 percent confidence intervals based on standard errors clustered by radio market using the leave-one-cluster-out adjustment. The vertical dashed line separates the pre-treatment and post-treatment periods.

D.3 Heterogeneity: IHS Robustness

Figure 18: Heterogeneity in Revenue Effects by Baseline Network-Dependence Quartile: IHS Specification



Notes: The figure reports Borusyak–Jaravel–Spiess (BJS) estimates of the effects of television entry on inverse hyperbolic sine transformed major-network revenue and local advertising revenue by quartile of baseline network dependence. The sample includes stations with positive baseline major-network revenue. Baseline network dependence is measured as the ratio of major-network revenue to time-sales revenue over 1946–1948. Vertical bars represent 95 percent confidence intervals. Standard errors are clustered by radio market.

D.4 Alternative Staggered-Adoption Estimator

As a robustness check to the Borusyak–Jaravel–Spiess estimates, I re-estimate the effects of television entry using the Callaway–Sant'Anna staggered difference-in-differences estimator. The estimator uses not-yet-treated markets as the comparison group and allows treatment effects to vary across treatment cohorts and calendar time. Appendix Table 13 reports the pooled estimates for the same revenue outcomes examined in the main analysis.

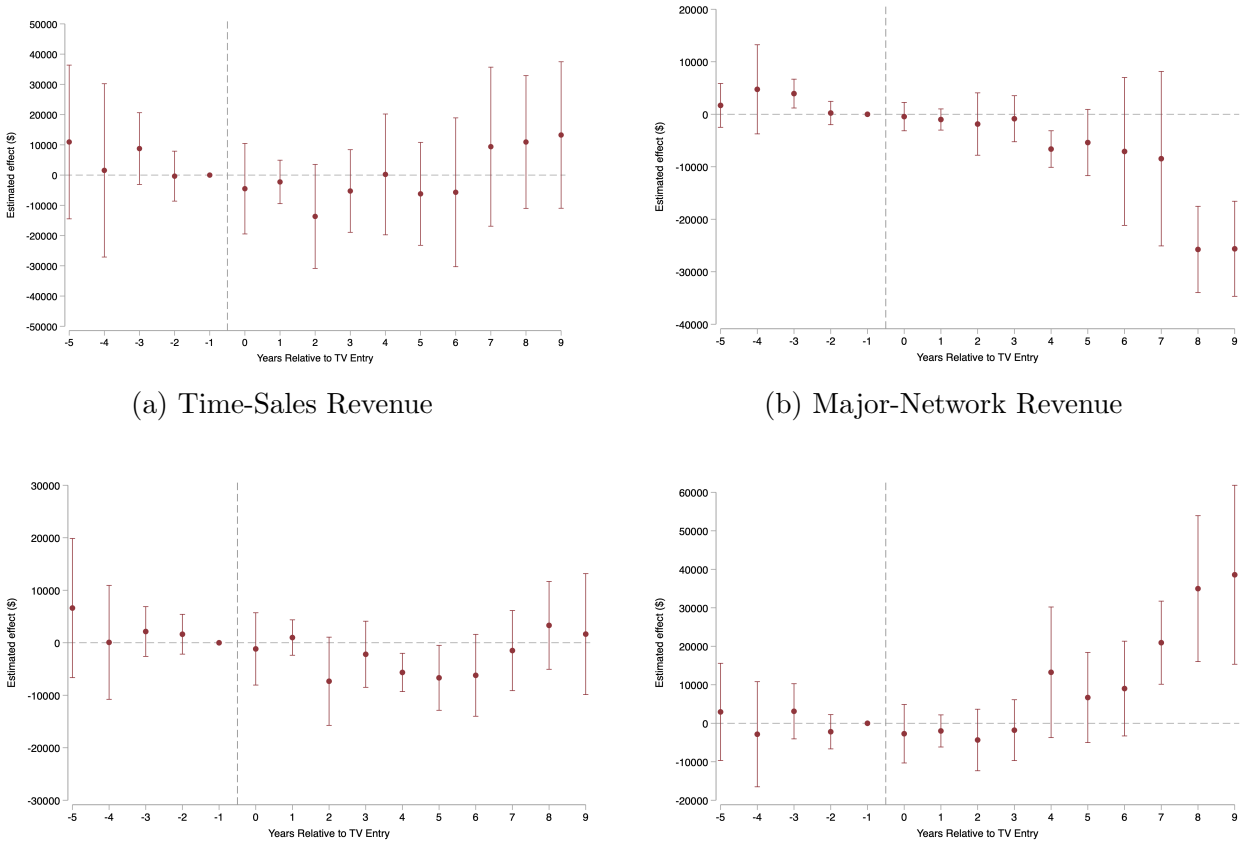
Table 13: Television Entry and Radio Revenue: Callaway–Sant'Anna Estimates

	Time-sales revenue (1)	Major-network revenue (2)	National/regional advertising (3)	Local advertising revenue (4)
Panel A: Levels (1950 $)				
Average treatment effect	3,294	-8,674***	-675	13,520***
	(6,687)	(2,912)	(2,197)	(4,848)
Joint pretrend p-value	0.529	0.008	0.743	0.669
Observations	7,224	7,224	7,224	7,224
Radio stations	858	858	858	858
Clusters	454	454	454	454
Panel B: IHS				
Average treatment effect	0.1163	-0.3471	-0.0766	0.2494**
	(0.0916)	(0.2299)	(0.1114)	(0.1042)
Joint pretrend p-value	0.463	0.310	0.349	0.369
Observations	7,224	7,224	7,224	7,224
Radio stations	858	858	858	858
Clusters	454	454	454	454
Radio station FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: This table reports average treatment effects estimated using the Callaway–Sant'Anna staggered difference-in-differences estimator for the freeze-based research-design sample. Treatment begins in the calendar year in which a radio market first receives television exposure, and not-yet-treated markets form the comparison group. Estimates use the doubly robust estimator and balance the panel separately within each underlying 2×2 comparison. With panel data, the underlying two-period difference-in-differences comparisons absorb radio-station and calendar-year effects; the table therefore reports radio-station and year fixed effects as included. Panel A reports revenue outcomes in constant 1950 U.S. dollars. Time-sales revenue includes payments from major, regional, and other networks as well as national/regional and local advertising revenue; column (2) isolates revenue received from the major national networks. Panel B reports inverse hyperbolic sine transformations of the same outcomes after expressing revenues in thousands of constant 1950 U.S. dollars. The joint pretrend p-value tests whether the pre-treatment event-study coefficients are jointly equal to zero. Standard errors are clustered by radio market. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Figure 19 reports the Callaway–Sant'Anna dynamic estimates in revenue levels, while Appendix Figure 20 reports the corresponding inverse hyperbolic sine specifications.

Figure 19: Dynamic Effects of Television Entry: Callaway–Sant'Anna Estimates



(a) Time-Sales Revenue

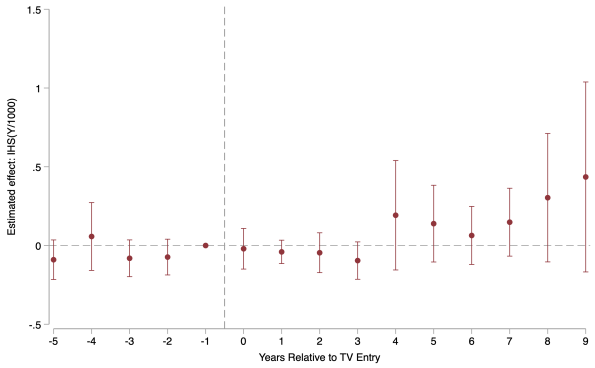
(b) Major-Network Revenue

(c) National/Regional Advertising

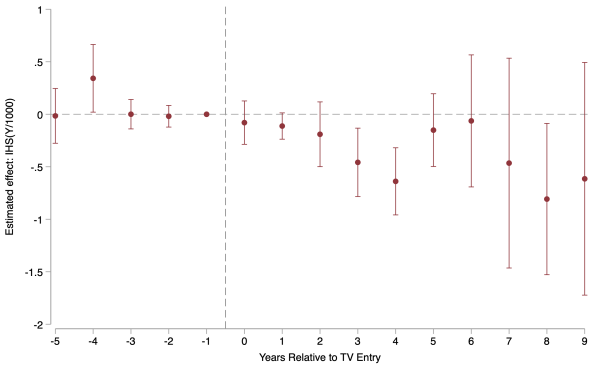
(d) Local Advertising

Notes: The figure reports event-study estimates using the Callaway–Sant'Anna staggered difference-in-differences estimator for the freeze-based research-design sample. Treatment begins in the calendar year in which a radio market first receives television exposure, and not-yet-treated markets serve as the comparison group. Estimates use the doubly robust estimator and balance the panel separately within each underlying 2×2 comparison. Event time -1 is normalized to zero. All panels report revenue outcomes in constant 1950 U.S. dollars. Time-sales revenue includes payments from major, regional, and other networks as well as national/regional and local advertising revenue. Vertical bars report 95 percent pointwise confidence intervals based on analytical standard errors clustered by radio market. The vertical dashed line separates the pre-treatment and post-treatment periods.

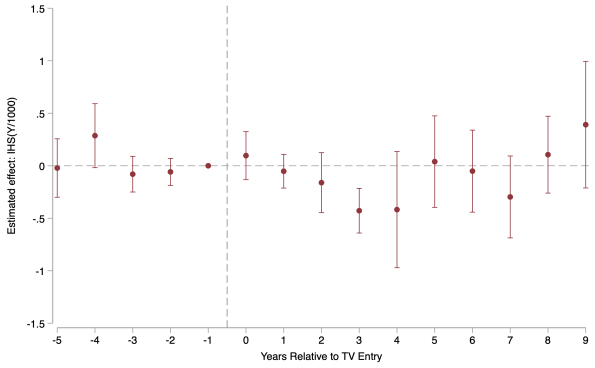
Figure 20: Dynamic Effects of Television Entry: Callaway–Sant'Anna Estimates, IHS



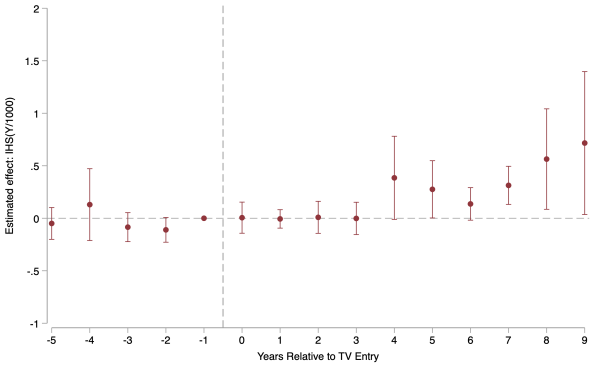
(a) Time-Sales Revenue



(b) Major-Network Revenue



(c) National/Regional Advertising



(d) Local Advertising

Notes: The figure reports the inverse hyperbolic sine counterparts to the Callaway–Sant'Anna event-study estimates shown in Appendix Figure 19. Monetary outcomes are expressed in thousands of constant 1950 U.S. dollars before applying the inverse hyperbolic sine transformation. Treatment begins in the calendar year in which a radio market first receives television exposure, and not-yet-treated markets serve as the comparison group. Estimates use the doubly robust estimator and balance the panel separately within each underlying 2×2 comparison. Event time -1 is normalized to zero. Vertical bars report 95 percent pointwise confidence intervals based on analytical standard errors clustered by radio market. The vertical dashed line separates the pre-treatment and post-treatment periods.

D.5 Baseline Major-Network Revenue Share: Alternative Estimator

Appendix Table 14 examines heterogeneity by baseline major-network revenue share using the Callaway–Sant'Anna estimator. These estimates provide an alternative-estimator robustness check for the heterogeneity results.

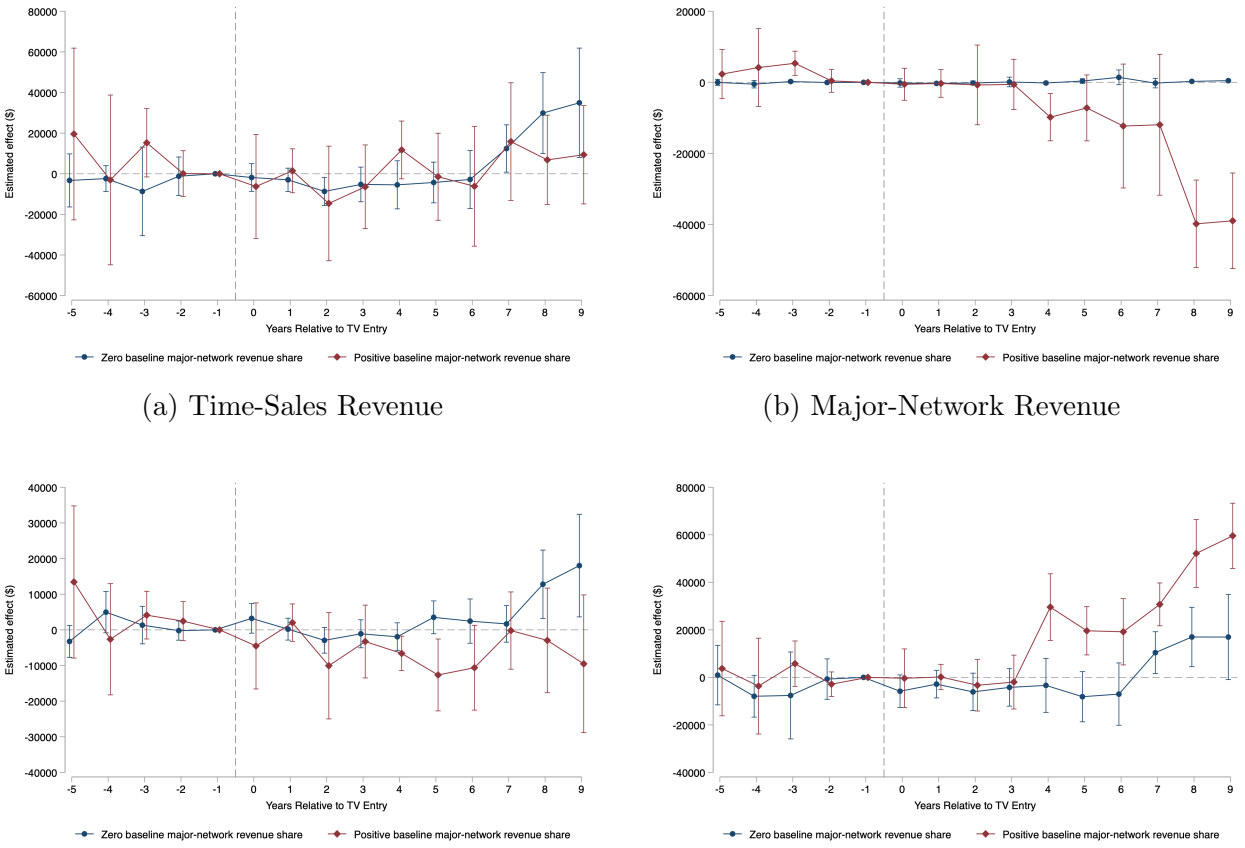
Table 14: Television Entry by Baseline Major-Network Revenue Share: Callaway–Sant'Anna Estimates

	Time-sales revenue (1)	Major-network revenue (2)	National/regional advertising (3)	Local advertising revenue (4)
Panel A: Levels (1950 $)				
Zero baseline major-network share	7,354	155	4,435*	2,691
	(5,038)	(246)	(2,323)	(3,889)
Positive baseline major-network share	5,939	-12,666***	-3,507	23,525***
	(6,975)	(4,053)	(3,318)	(3,789)
Joint pretrend p-value: Zero	0.808	0.015	0.228	0.344
Joint pretrend p-value: Positive	0.195	0.004	0.611	0.252
Panel B: IHS				
Zero baseline major-network share	0.0033	-0.0638	0.2140	0.0672
	(0.0561)	(0.0918)	(0.1360)	(0.0736)
Positive baseline major-network share	0.2570**	-0.1780	-0.0068	0.4350***
	(0.1110)	(0.3160)	(0.1090)	(0.1180)
Joint pretrend p-value: Zero	0.726	0.001	0.475	0.519
Joint pretrend p-value: Positive	0.659	0.346	0.503	0.223
Zero baseline-share sample	2,362 observations; 286 radio stations; 254 clusters			
Positive baseline-share sample	4,331 observations; 454 radio stations; 266 clusters			
Radio station FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: This table reports heterogeneous treatment effects estimated separately by baseline major-network revenue share using the Callaway–Sant'Anna staggered difference-in-differences estimator for the freeze-based research-design sample. Baseline share is measured over 1946–1948 as the ratio of major-network revenue to time-sales revenue across usable baseline observations; stations are classified into zero-share and strictly positive-share groups. Treatment begins in the first calendar year of local television exposure, with not-yet-treated markets as the comparison group. Estimates use the doubly robust estimator and balance the panel separately within each underlying 2×2 comparison. With panel data, these comparisons absorb radio-station and calendar-year effects, so the table reports radio-station and year fixed effects as included. Panel A reports revenues in constant 1950 U.S. dollars; Panel B reports inverse hyperbolic sine transformations after expressing revenues in thousands of constant 1950 U.S. dollars. Time-sales revenue includes network payments and national/regional and local advertising revenue; column (2) isolates major-network revenue. Subgroup pretrend p-values jointly test the corresponding pre-treatment event-study coefficients. Because subgroup effects are estimated separately, no formal cross-group equality test is reported. Standard errors are clustered by radio market. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Figure 21 reports the heterogeneous Callaway–Sant'Anna dynamic estimates in revenue levels, while Appendix Figure 22 reports the corresponding inverse hyperbolic sine specifications.

Figure 21: Dynamic Effects of Television Entry by Baseline Major-Network Revenue Share: Callaway–Sant'Anna Estimates



(a) Time-Sales Revenue

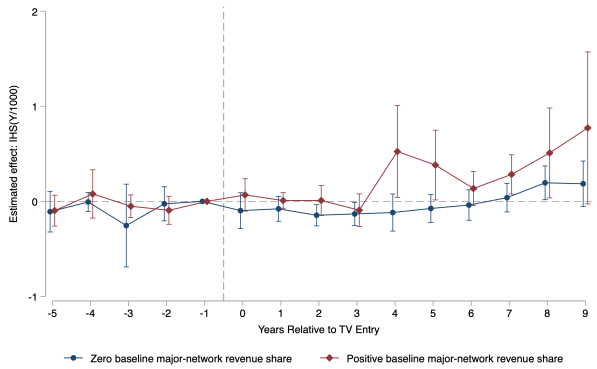
(b) Major-Network Revenue

(c) National/Regional Advertising

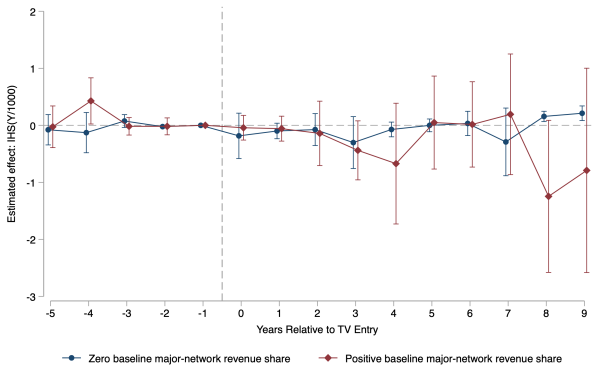
(d) Local Advertising

Notes: The figure reports event-study estimates using the Callaway–Sant'Anna staggered difference-in-differences estimator, estimated separately for stations with zero and strictly positive baseline major-network revenue share in the freeze-based research-design sample. Baseline major-network revenue share is measured over 1946–1948 as the ratio of major-network revenue to time-sales revenue across usable baseline observations. Treatment begins in the calendar year in which a radio market first receives television exposure, and not-yet-treated markets serve as the comparison group. Estimates use the doubly robust estimator and balance the panel separately within each underlying 2×2 comparison. Event time -1 is normalized to zero. All panels report revenue outcomes in constant 1950 U.S. dollars. Time-sales revenue includes payments from major, regional, and other networks as well as national/regional and local advertising revenue. Vertical bars report 95 percent pointwise confidence intervals based on analytical standard errors clustered by radio market. The vertical dashed line separates the pre-treatment and post-treatment periods.

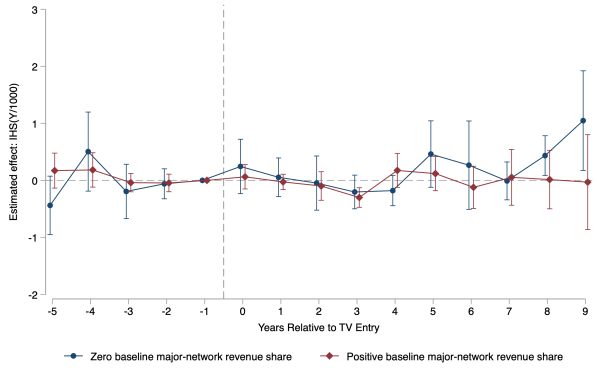
Figure 22: Dynamic Effects of Television Entry by Baseline Major-Network Revenue Share: Callaway–Sant'Anna Estimates, IHS



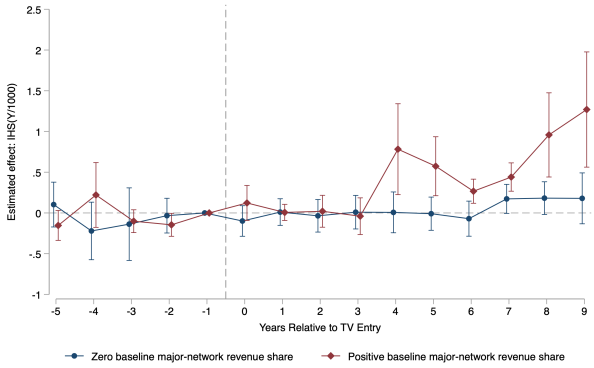
(a) Time-Sales Revenue



(b) Major-Network Revenue



(c) National/Regional Advertising



(d) Local Advertising

Notes: The figure reports the inverse hyperbolic sine counterparts to the heterogeneous Callaway–Sant'Anna event-study estimates shown in Appendix Figure 21. Estimates are reported separately for stations with zero and strictly positive baseline major-network revenue share in the freeze-based research-design sample. Baseline major-network revenue share is measured over 1946–1948 as the ratio of major-network revenue to time-sales revenue across usable baseline observations. Monetary outcomes are expressed in thousands of constant 1950 U.S. dollars before applying the inverse hyperbolic sine transformation. Treatment begins in the calendar year in which a radio market first receives television exposure, and not-yet-treated markets serve as the comparison group. Estimates use the doubly robust estimator and balance the panel separately within each underlying 2×2 comparison. Event time -1 is normalized to zero. Vertical bars report 95 percent pointwise confidence intervals based on analytical standard errors clustered by radio market. The vertical dashed line separates the pre-treatment and post-treatment periods.

D.6 Television Entry and the Number of Observed Radio Stations

One possible explanation for the increase in local advertising revenue is that television altered the number of radio stations operating in a market, thereby changing the competitive environment facing incumbent stations. As a supplementary check, I examine whether television entry changes the number of radio stations observed in each radio market. This outcome counts all radio stations observed in the American Radio Panel in a given market-year, rather than restricting the count to the incumbent stations used in the main revenue analysis.

Appendix Table 15 reports estimates using both the Borusyak–Jaravel–Spiess imputation estimator and the Callaway–Sant'Anna estimator. The estimated effects are small and statistically indistinguishable from zero under both estimators. I therefore find no detectable change in the number of observed radio stations in these markets over the period studied. The result is consistent with the main short-run adjustment appearing within incumbent stations rather than in the number of radio stations observed in the market.

Table 15: Television Entry and the Number of Observed Radio Stations

	BJS	Callaway–Sant'Anna
	(1)	(2)
Effect of TV entry	-0.076	-0.158
	(0.223)	(0.248)
Untreated mean	1.734	1.734
Joint pretrend p-value	0.145	0.404
Market-year observations	4,662	4,662
Markets	444	444

Notes: The dependent variable is the number of radio stations observed in a radio market in each reporting year. The count includes all radio stations observed in the American Radio Panel in the 444 markets represented in the preferred BJS revenue specification and is therefore not restricted to the pre-freeze incumbent stations used as outcomes in the main station-level analysis. Column (1) reports the treatment effect estimated using the Borusyak–Jaravel–Spiess imputation estimator, and column (2) reports the corresponding estimate using the Callaway–Sant'Anna staggered difference-in-differences estimator. Treatment begins in the calendar year in which a market first receives television exposure. Standard errors, reported in parentheses, are clustered by radio market. The joint pretrend p-value tests the corresponding pre-treatment coefficients for each estimator.